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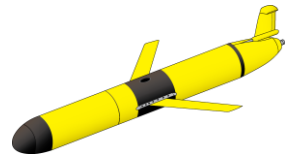
Teledyne Webb Research
Slocum Glider Training



FAL-TG-21-004

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Slocum Glider – *The Basics*



- Introductions
- Agenda for the week
- Glider Overview
  - Glider Components
  - Glider Configurations
  - Glider Flight Fundamentals
  - Glider Sensors
  - Product Applications



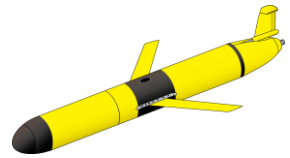
Photo: Tiffany Wardman

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The Slocum Glider



What is the Slocum Glider?

Buoyancy engine driven, ultra long-endurance AUVs for persistent ocean monitoring

What problem does it solve?

- Persistent data collection at a fraction of the cost of surface ships or traditional AUVs
- Increased data points at a higher frequency than possible with traditional means.
- Less life cycle cost
- Safer, more cost effective and higher quality data than gathered through traditional means.

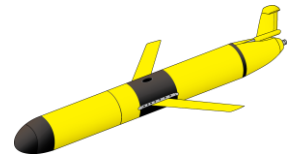


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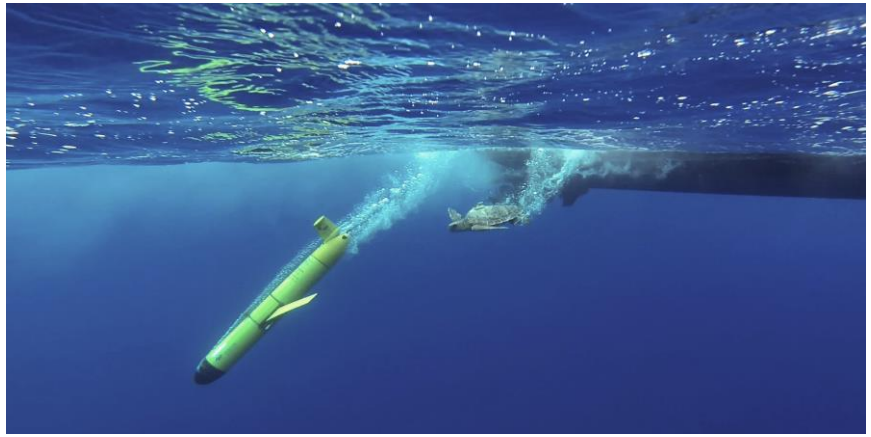
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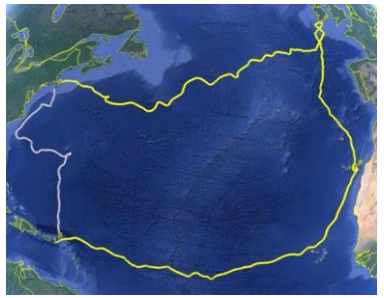
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Slocum Glider History



- Douglas Webb founded Webb Research in 1982 and developed the original Slocum Glider – named for Joshua Slocum, the first man to circumnavigate the globe solo in a sailboat
- Teledyne acquired Webb Research in 2008
- Slocum is the market leading glider – over 1000 delivered to Research, Academic, Military, and Commercial customers
- Proven performance:
  - In 2016-2020, Glider “Silbo” completed Atlantic Ocean circumnavigation with over 1,200 days at sea







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Slocum Glider History



Doug Webb has a <u>vision</u>

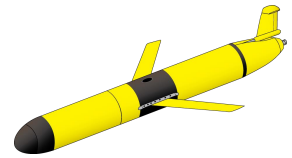
Work Hard, Have Fun, Change the World

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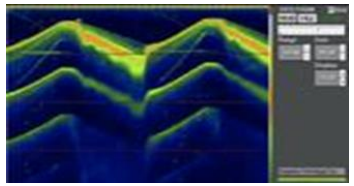
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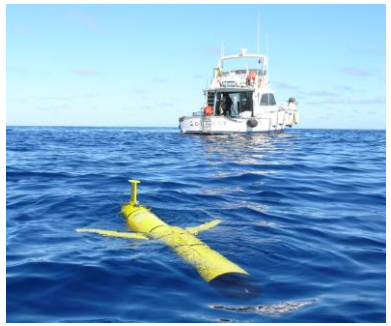
Slocum Product Applications



Science

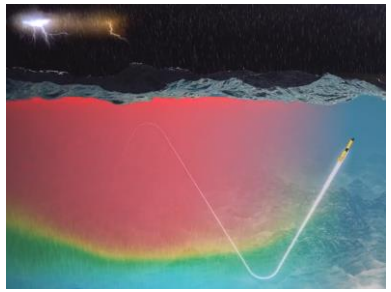
- Oceanography
- Data gateway
- Ocean acidity
- Storm Survey









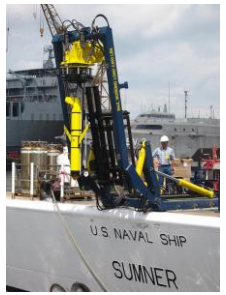


Defense

- Oceanographic data
- Specialty builds







Commercial

- Oil and Gas
- Offshore Wind
- Environmental monitoring including Current profiling
- Marine mammal mitigation
- Fisheries



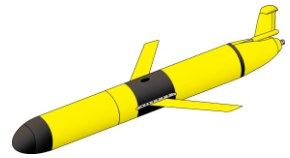




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Glider Support - *Philosophy & Structure*





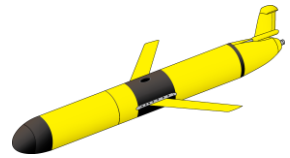
glidersupport@teledyne.com

- Wide distribution list including support, service, sales, engineering, and executive personnel
- Support ticketing system
- Often will be answered during off-hours if subject is compelling enough and if personnel are available

- The “Batphone” (for first deployment “emergencies only”)
  - (508) 524-8106
  - (855) 720-3915 (toll-free - US)



Glider Forum – *A great resource!*

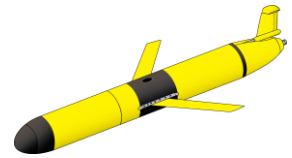


https://datahost.webbresearch.com/

- Most of the information we will discuss during this training is available on the Datahost, our Glider User Forum
- On the forum, you will be able to access:
  - Posts from fellow users & TWR employees on the use of gliders
  - Manuals
  - Software Releases
  - Other Glider Resources
  - Client Tools/SFMC
- Register if you have not already!
  - Once registered, contact Cordielyn.Goodrich@teledyne.com to have your account approved

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Glider Forum – *A great resource!*



TWR Forum
Slocum Glider support and ideas

Search... Search
Advanced search

Board index

User Control Panel (0 new messages) • View your posts — FAQ Members Logout [cgoodrich]

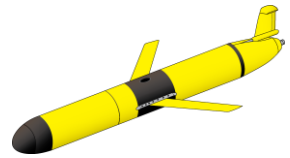
It is currently September 5th, 2021, 7:00 am — Last visit was: September 1st, 2021, 6:30 am

View unanswered posts • View unread posts • View new posts • View active topics — Mark forums read

GLIDERS	TOPICS	POSTS	LAST POST
G3S New SM32 Processor The Persistor replacement	5	6	by admin March 1st, 2021, 1:32 pm
FAQ Look here first.	94	240	by admin August 10th, 2021, 5:37 am
Operation Things to do with a glider in a boat	66	201	by stats August 10th, 2021, 3:55 pm
Missions Mission development	65	187	by Jude_Vdm July 26th, 2021, 11:47 am
lab testing	44	114	by admin July 22nd, 2021, 6:13 am
Suggestions	28	107	by ktikka May 8th, 2019, 10:26 pm
Aborts	17	54	by Jude_Vdm July 28th, 2021, 10:49 am
NEWS!!! Teledyne Webb Research news and share stories of your programs accomplishments!	13	15	by admin March 17th, 2021, 6:24 am
Dockserver post all GMC questions here	92	305	by admin July 23rd, 2021, 5:02 am
Resources Links to online glider resources	2	2	by admin May 2nd, 2014, 11:15 am
Glider Service Bulletins	1	1	by admin April 9th, 2014, 10:27 am

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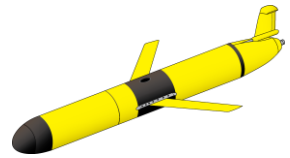
Objectives of Glider Training



- A trainee should be comfortable configuring & preparing a glider for a qualification or test flight
- Answer glider prep, maintenance, and piloting questions for all users
- We recommend starting slow and building a base of comfort and knowledge



Glider Basics



- Glider components
- Glider flight fundamentals
- Glider communication
- Shore side software - SFMC
- On-board glider software hierarchy



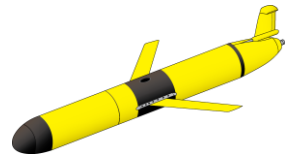
Photo: Tiffany Wardman

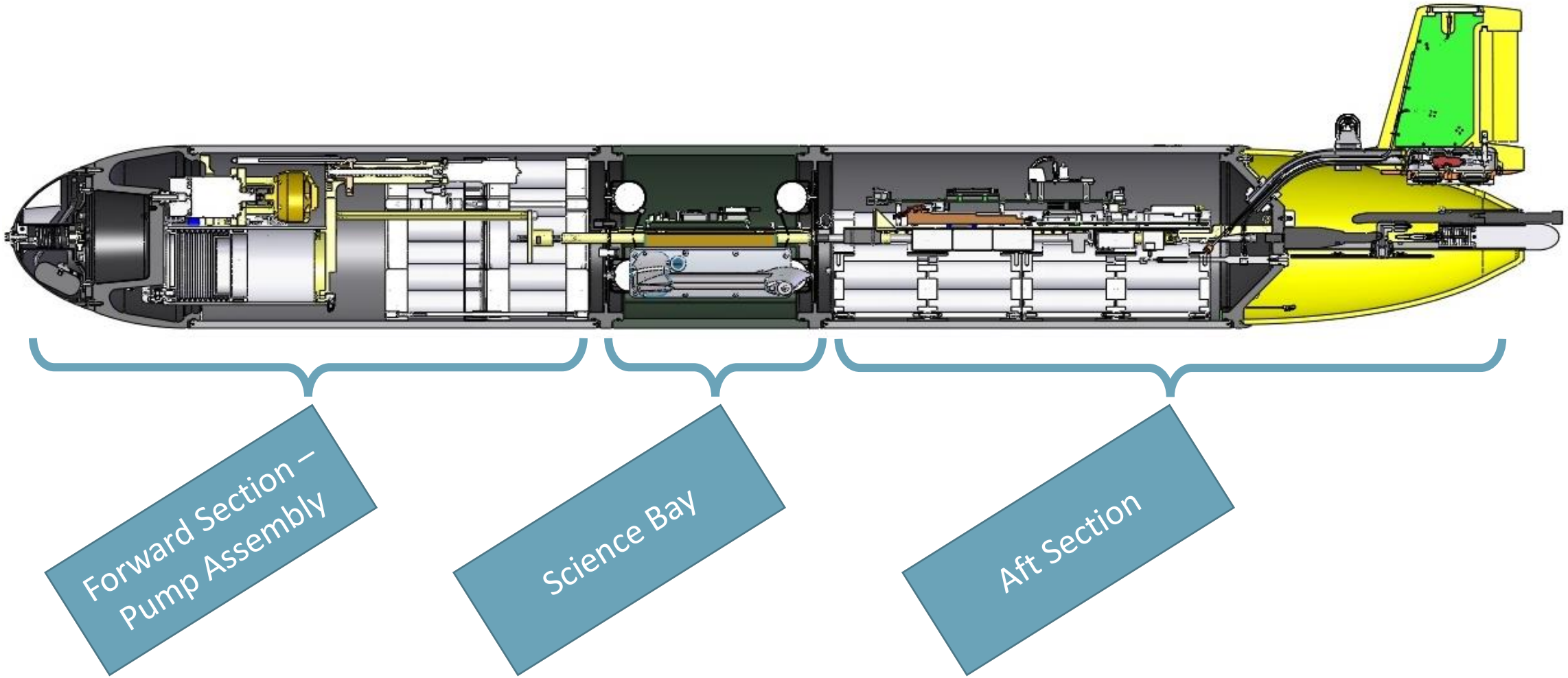
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Slocum Glider Components – Modularity



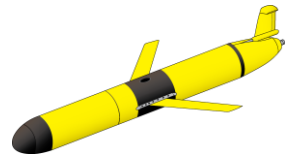


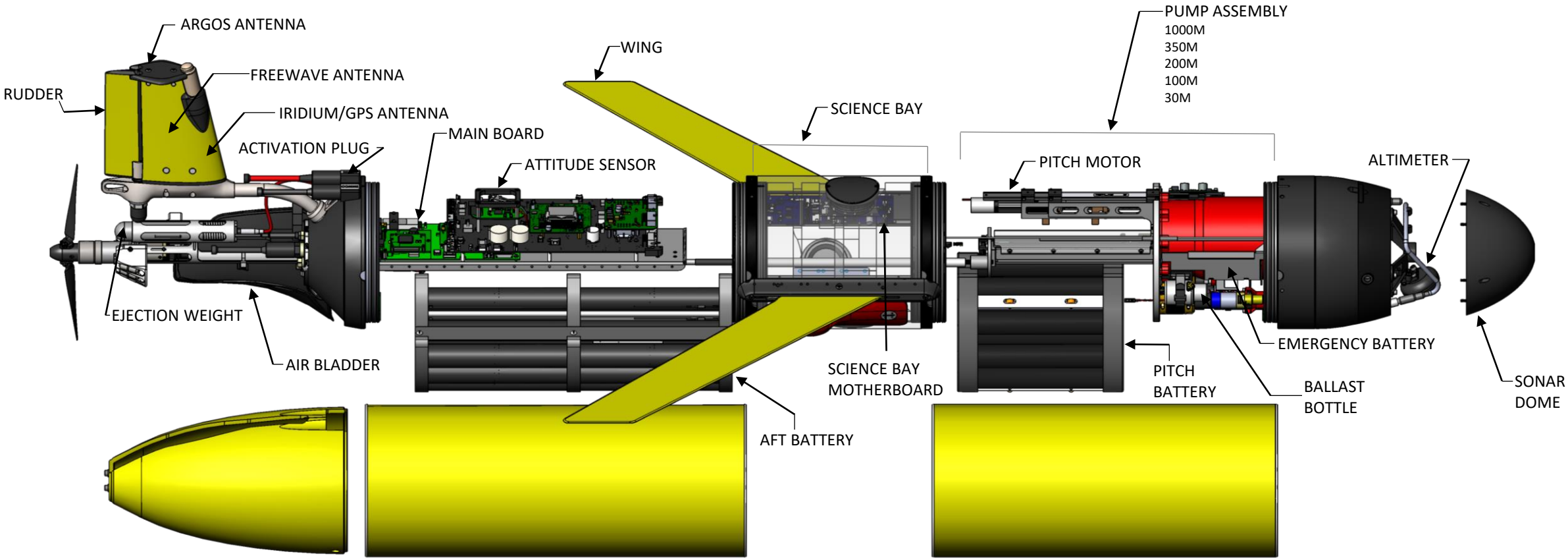
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Slocum Glider Components – G3



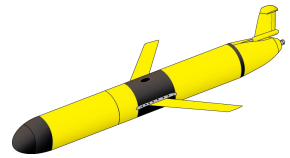


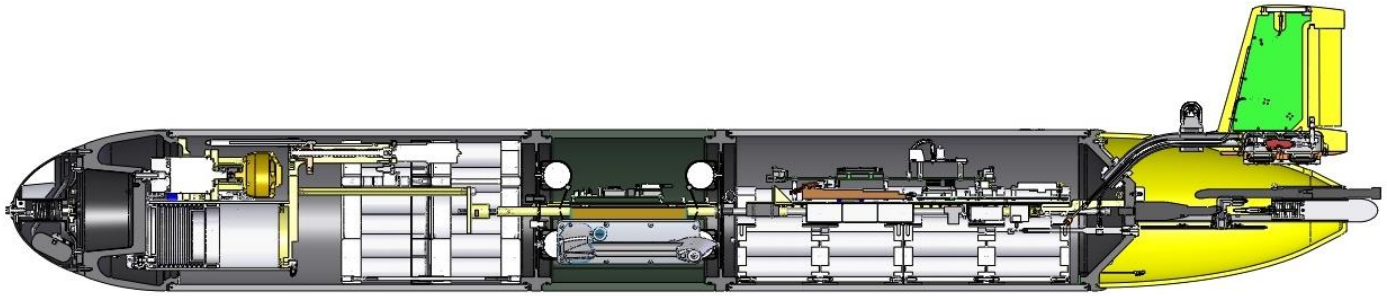
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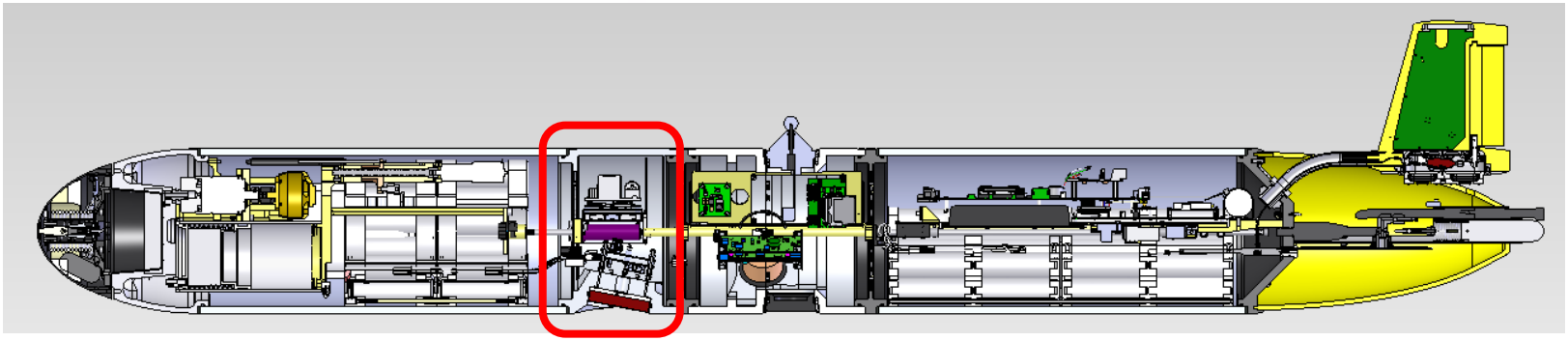
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Slocum Glider Configurations

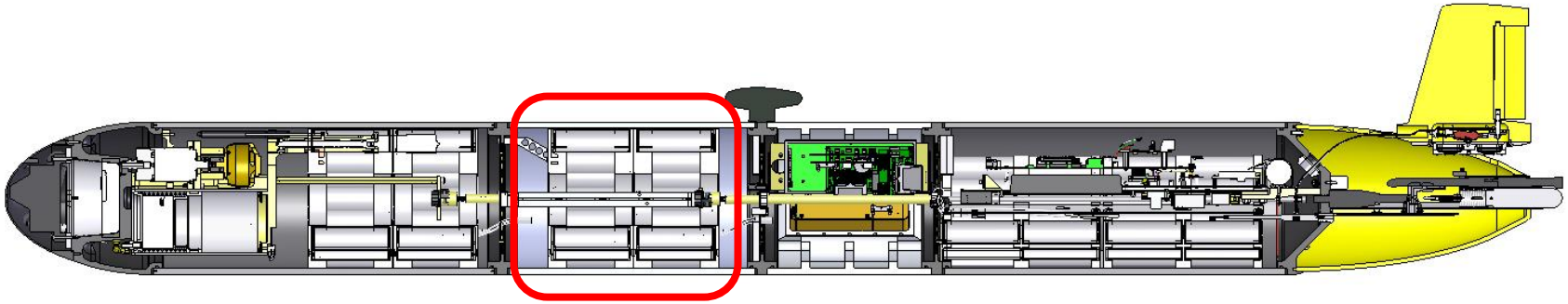




Standard Length Glider



Extended Glider with
Additional Sensor Bay



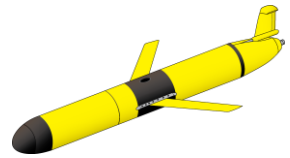
Extended Glider with
Extended Endurance Bay

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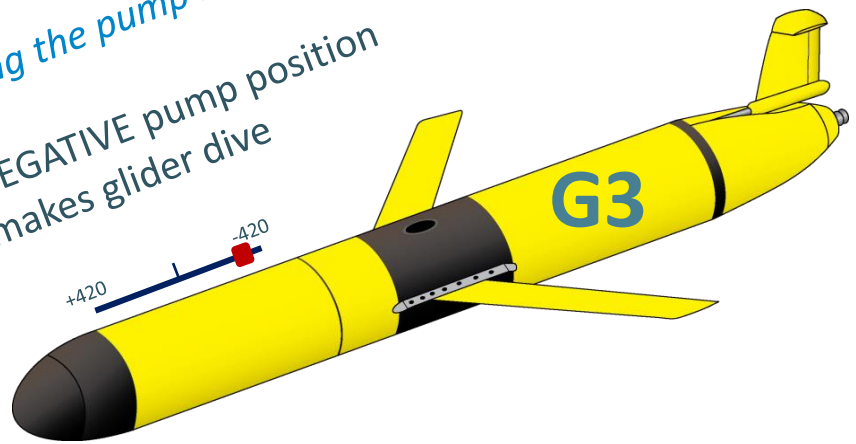
Buoyancy Pump Positions



Changing the position of the pump changes the volume of the vehicle (Pump position is measured in cc's)

Retracting the pump makes the glider more dense

NEGATIVE pump position
makes glider dive



POSITIVE pump position
makes glider climb



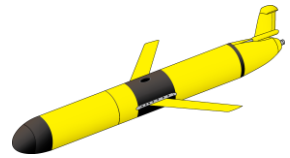
Extending the pump makes the glider less dense

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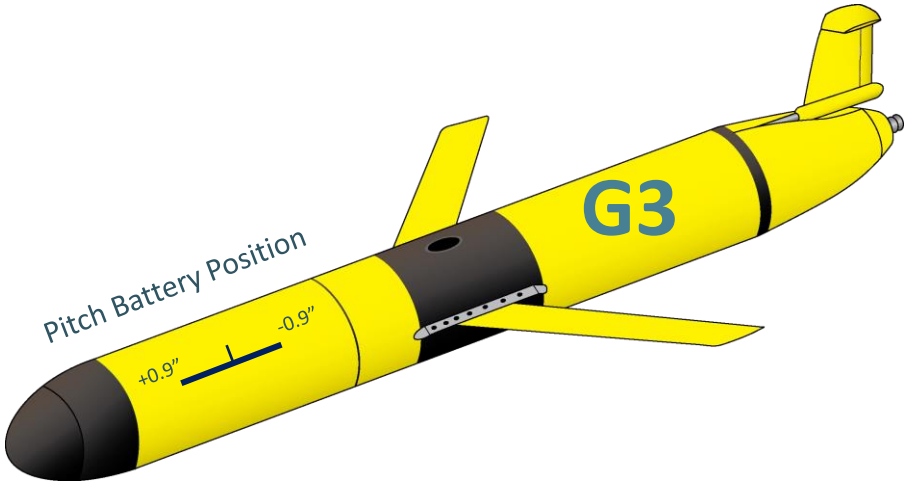
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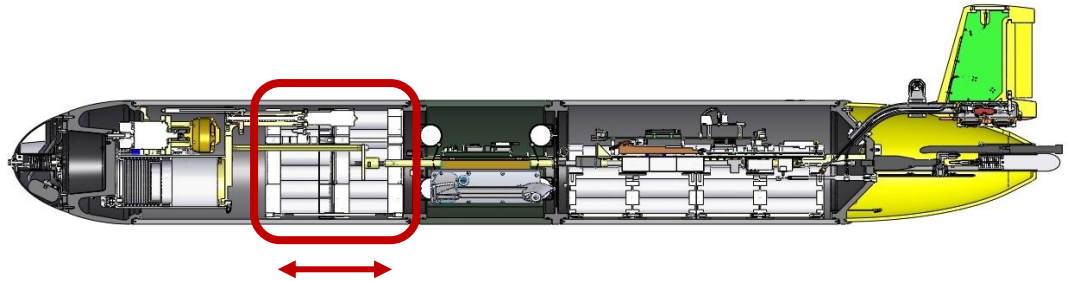
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Pitch Battery Positions



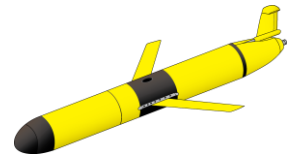
- Pitch battery position is measured in inches
- The pitch battery moves forward and aft to help the glider achieve the desired pitch angle
- Moving the pitch battery does not change the volume of vehicle!

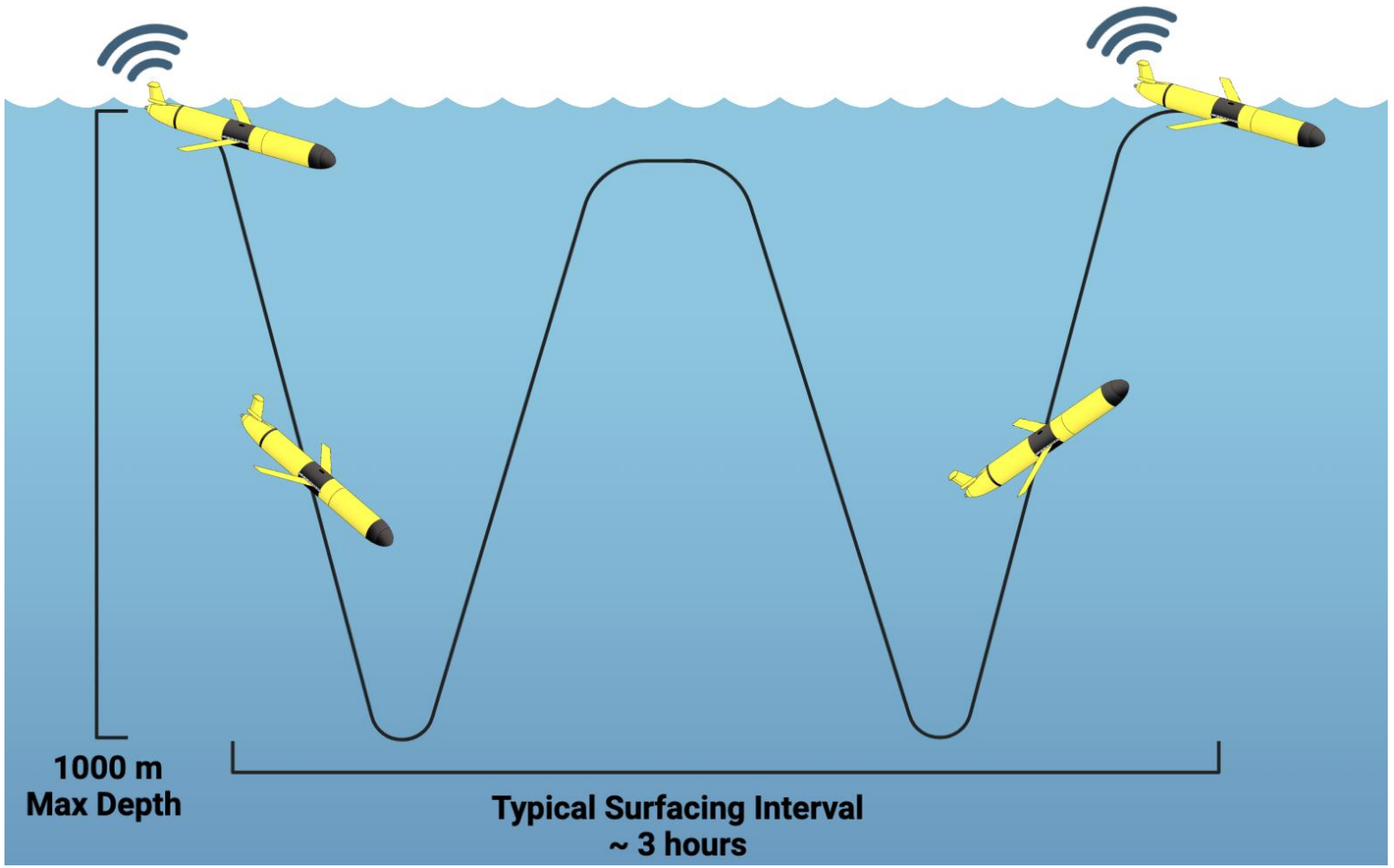




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Glider Flight Fundamentals





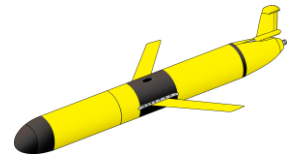
- Glider receives mission parameters
- Glider adjusts buoyancy and weight distribution to dive and surface, travelling in saw-tooth pattern
- Glider completes directed numbers of dives/surfaces (“yos”)
- Glider resurfaces, establishes GPS/comms, receives new mission parameters, transmits gathered data

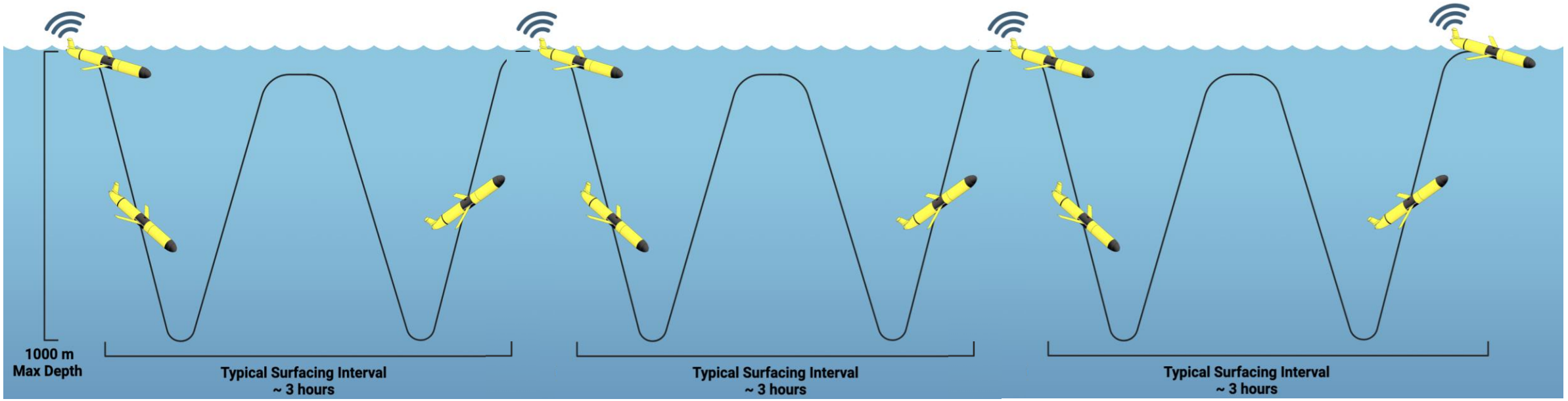
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Rhythm of a Glider Deployment

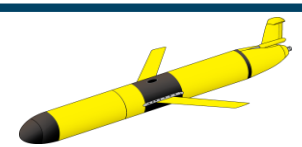


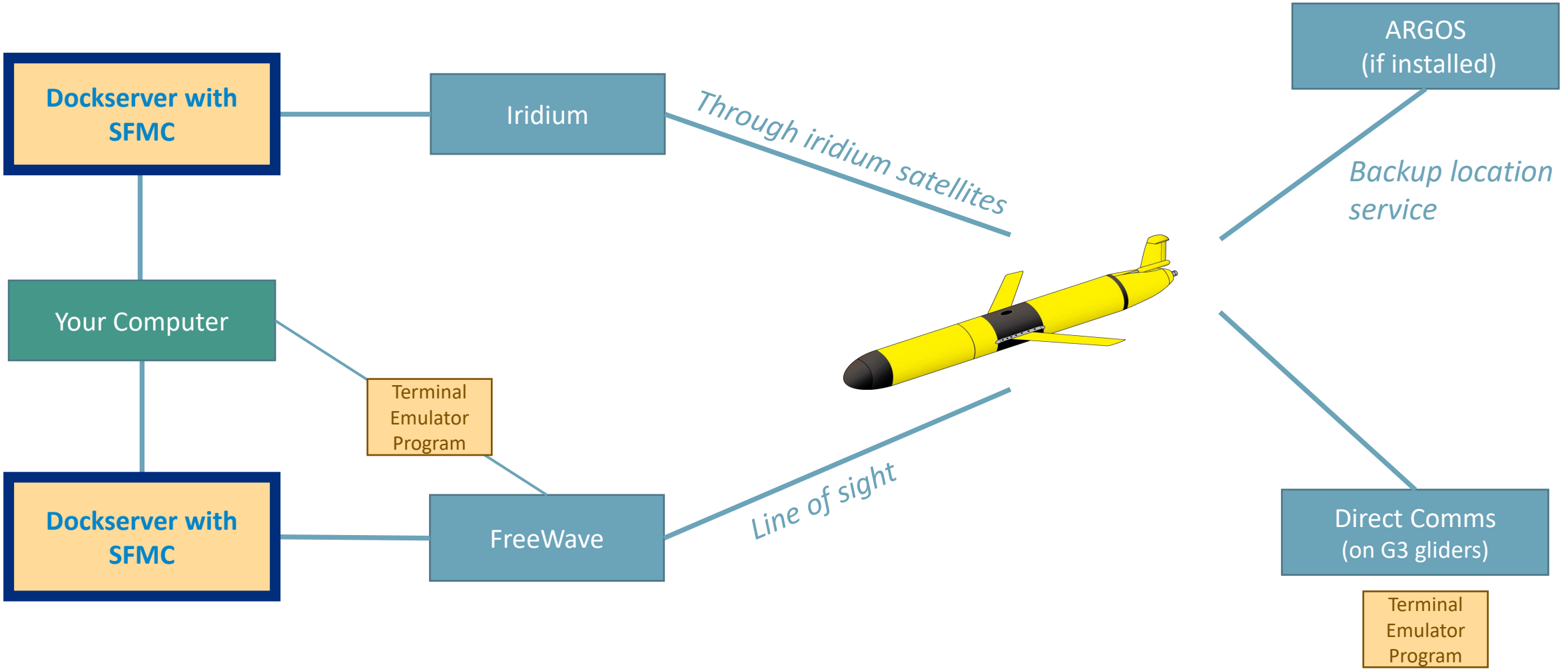


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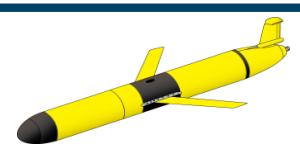
Glider Communication

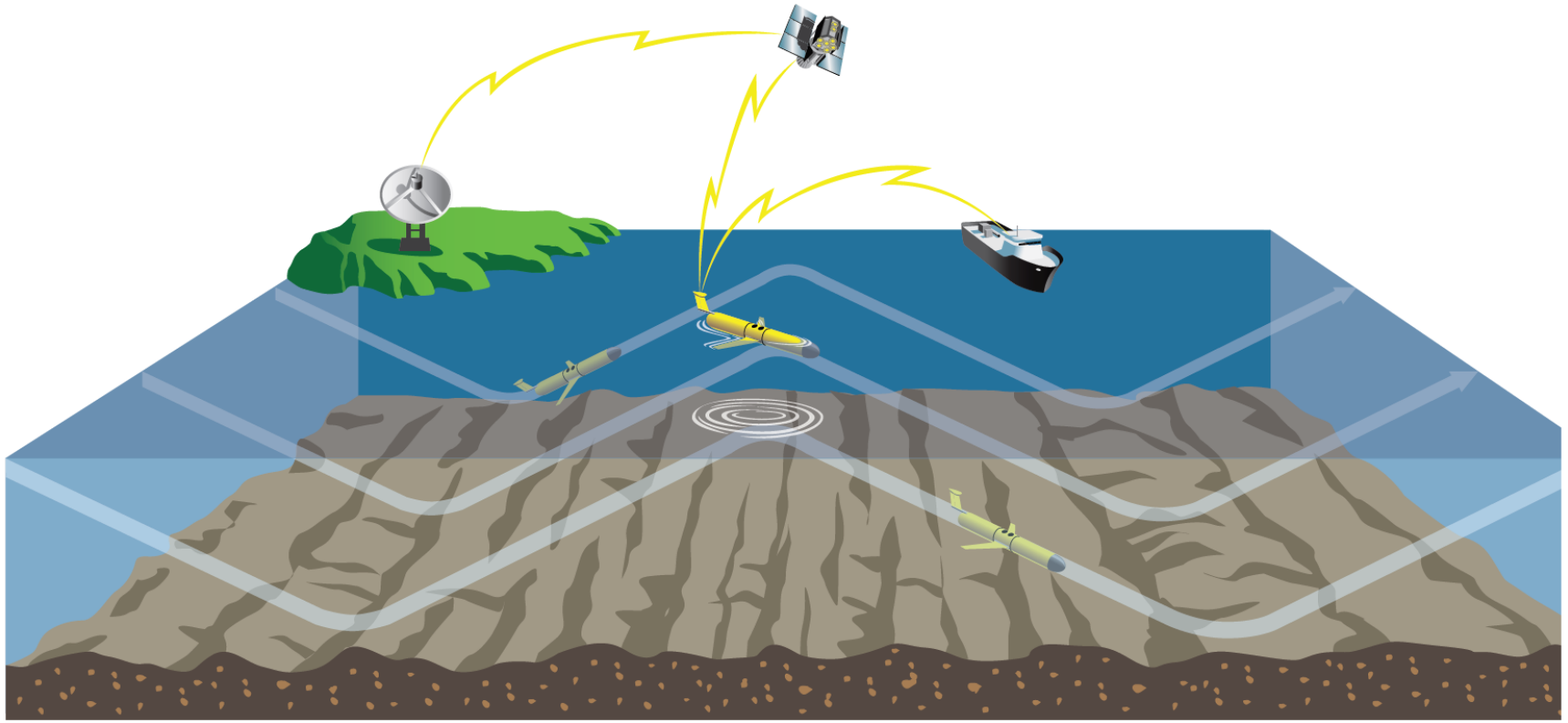




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Glider Communication



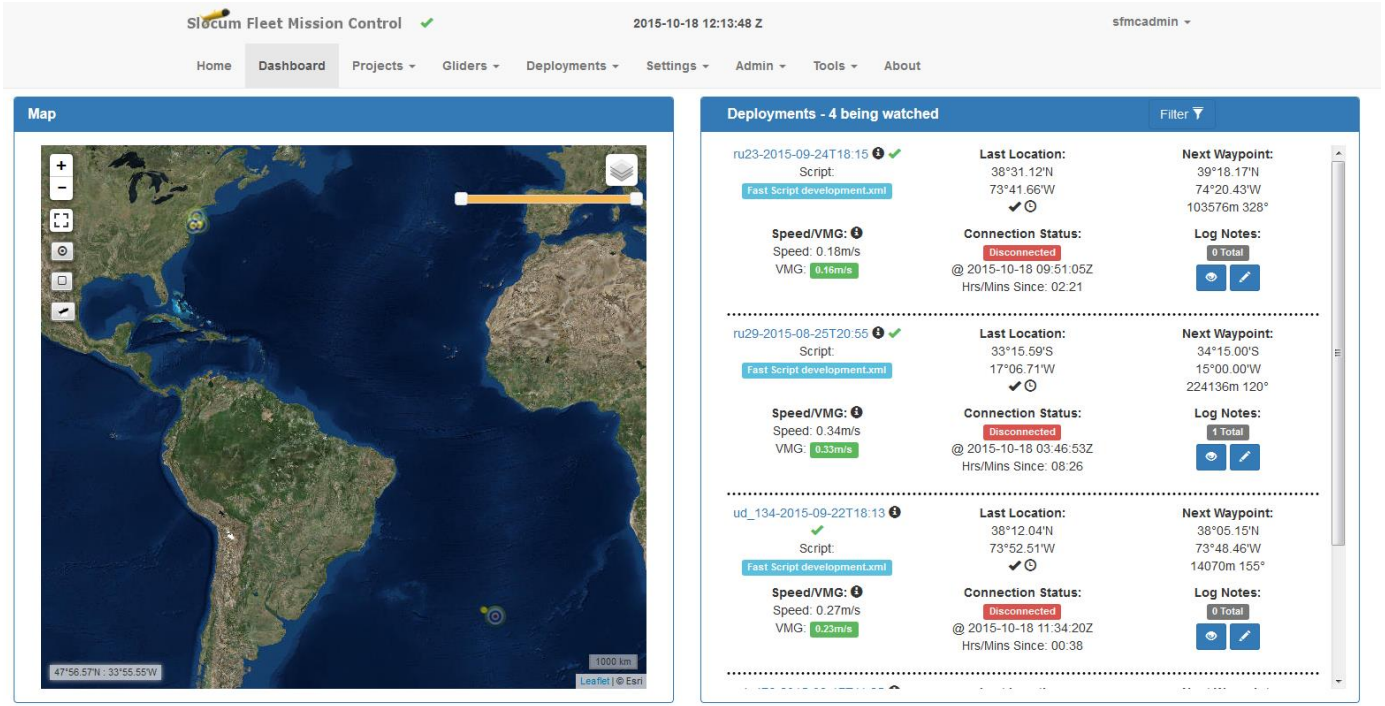


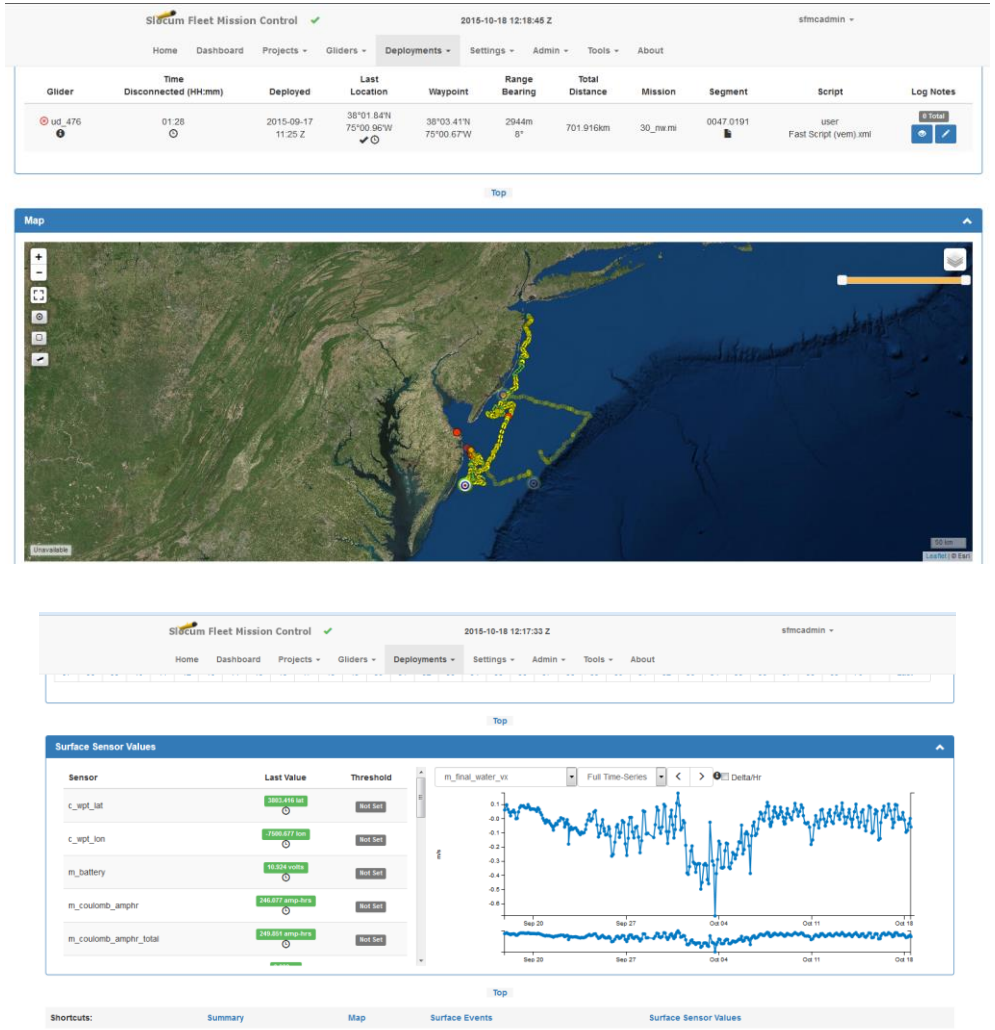
- Glider can communicate via Freewave comms any line-of-sight distance (up to 3 km)
- Glider can communicate via Iridium satellite to anywhere in the world

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Glider Software: Slocum Fleet Mission Control (SFMC)

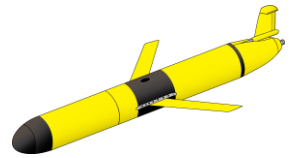




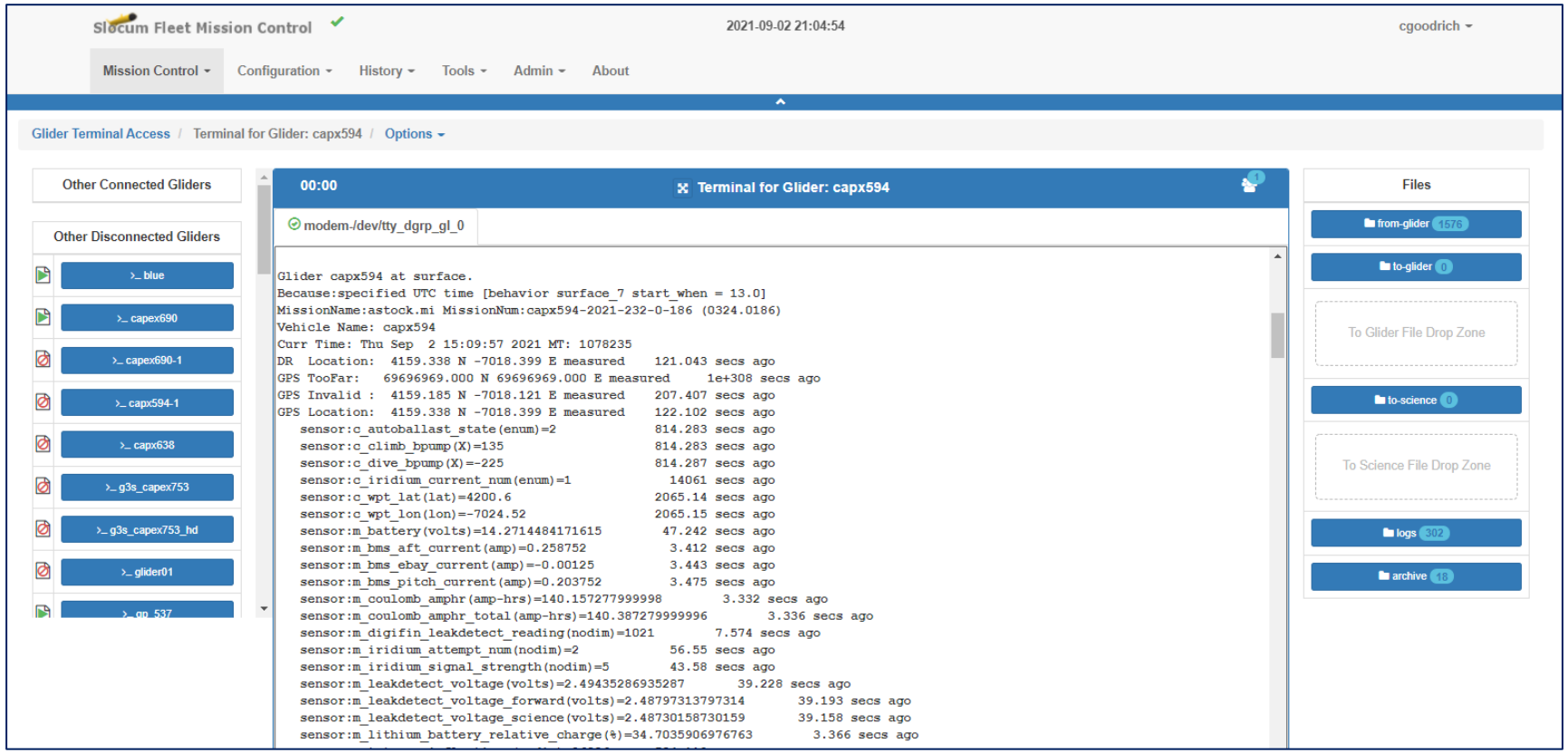
Comprehensive Slocum Fleet Mission Control (SFMC) software suite allows glider tracking, mission planning, and piloting from around the world

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Slocum Fleet Mission Control (SFMC)

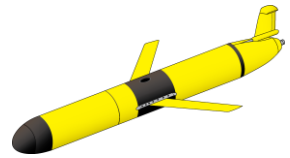


Glider Terminal

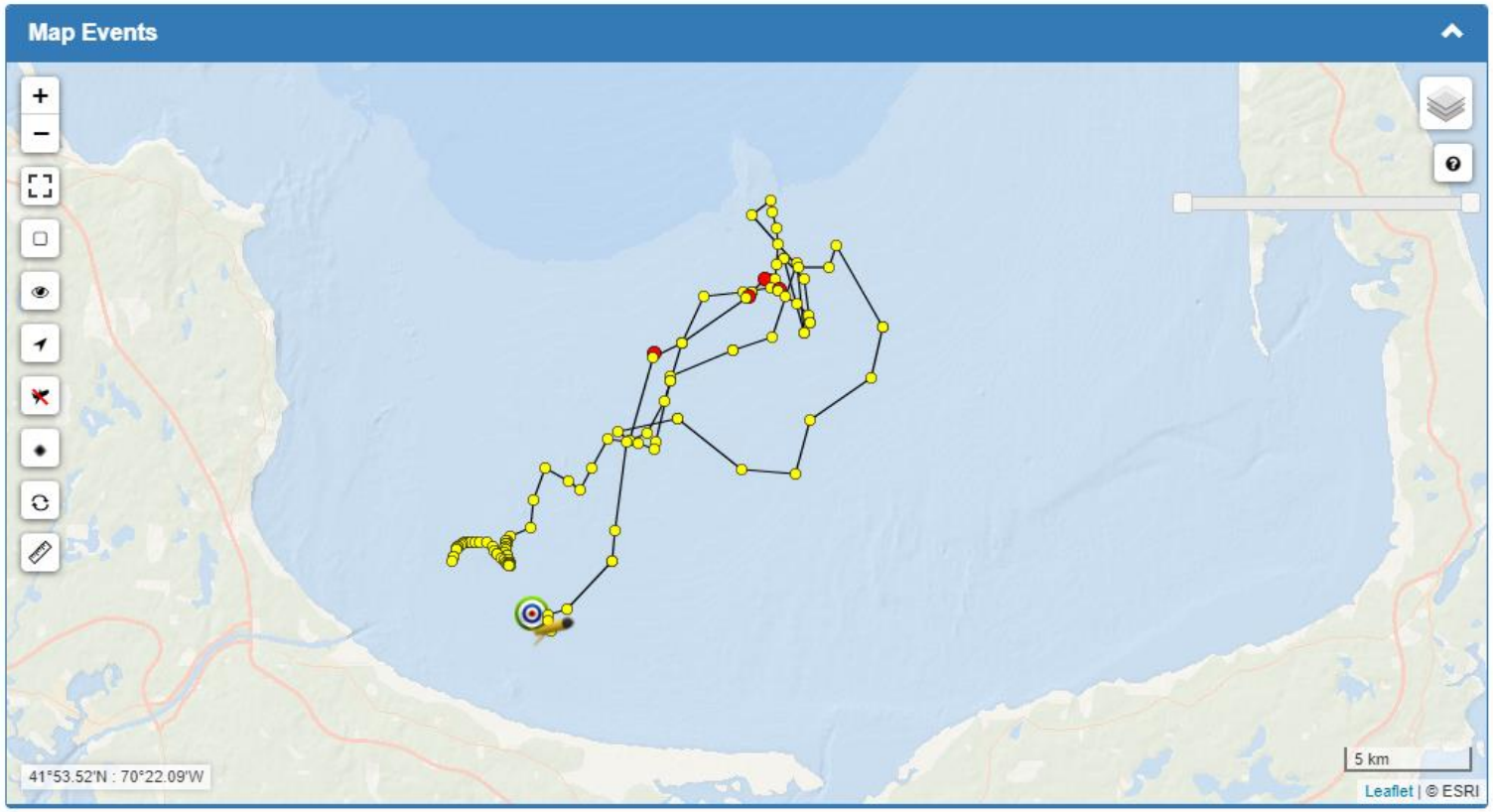


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Slocum Fleet Mission Control (SFMC)



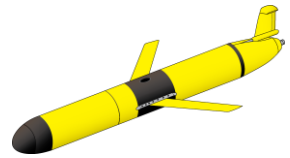
Deployment Maps



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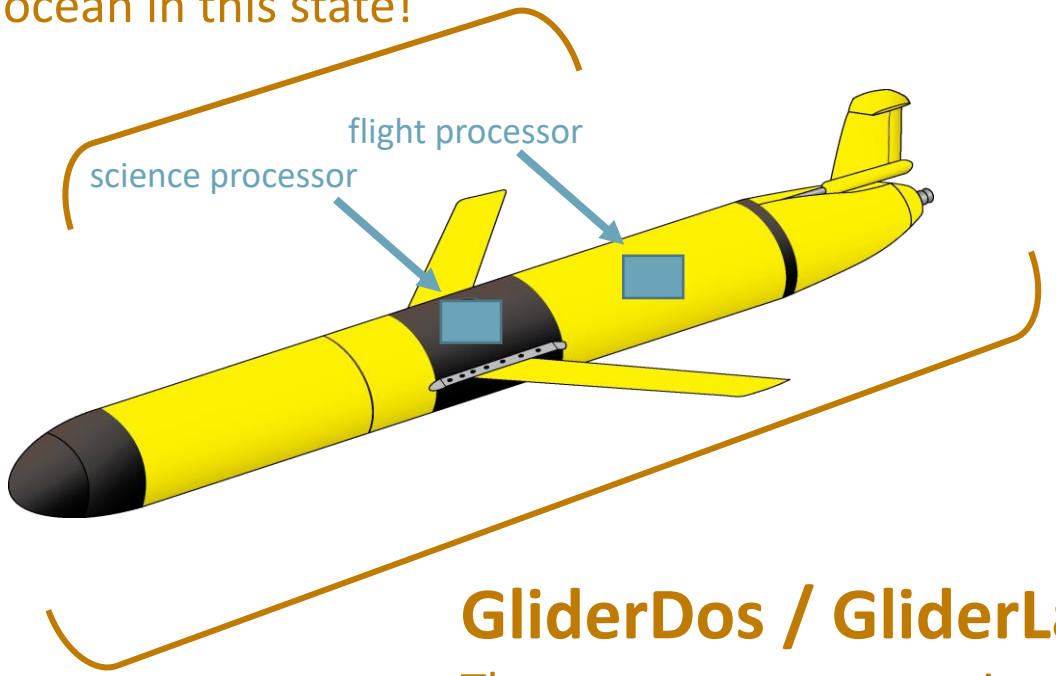
Glider Operating States



Pico (persistor based glider) / **Shell** (stm32 based glider)

Only talking to the processors (base OS)

You do not want to be in the ocean in this state!



GliderDos / GliderLab

The processors are running the glider application

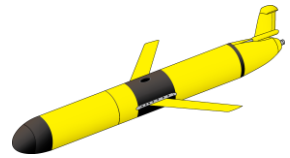
The glider can use important devices here

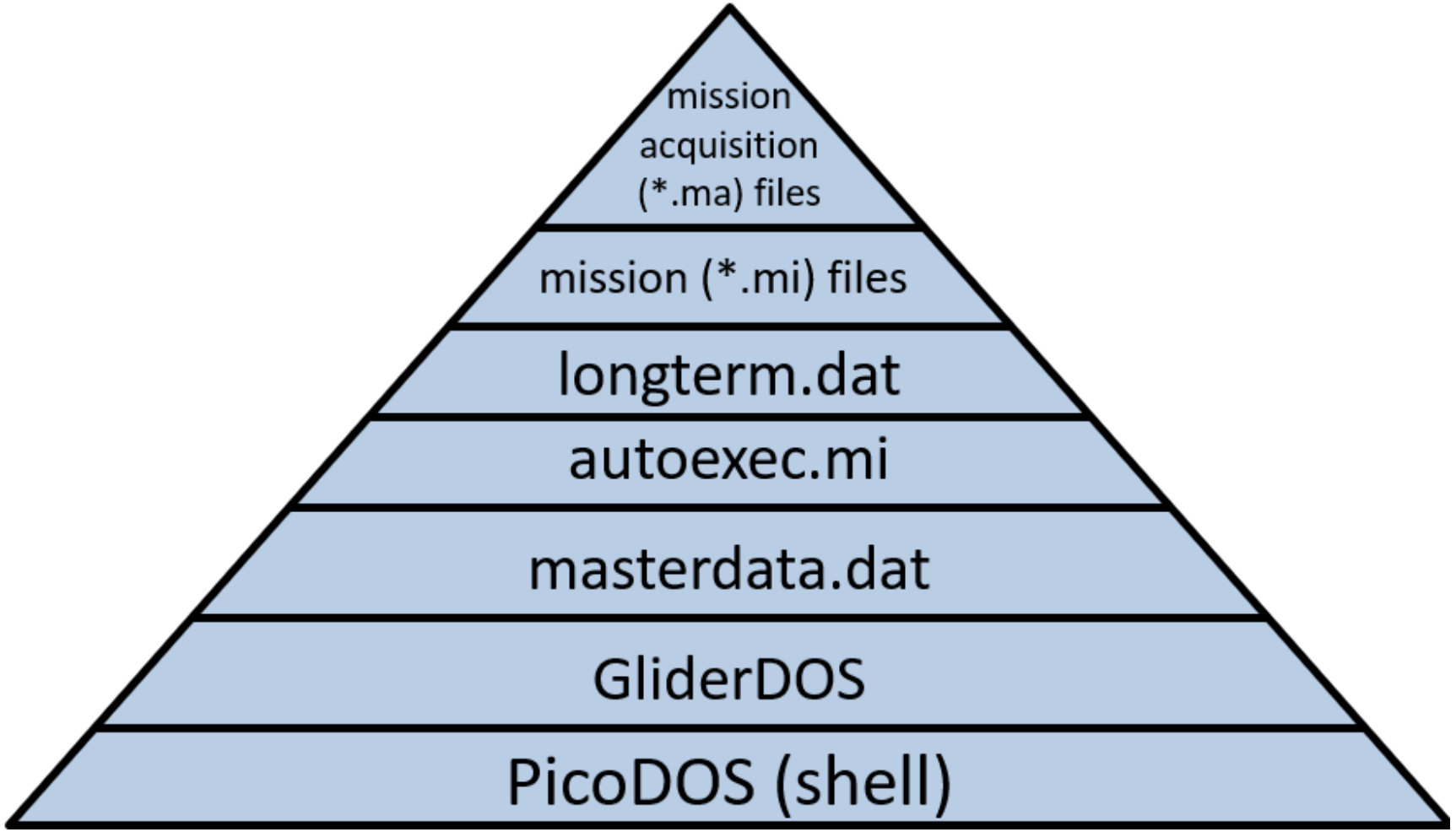
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Glider Software – *Hierarchy of Glider Software Needs*



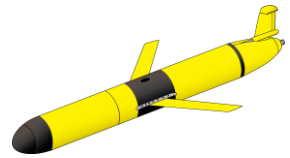


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Glider Processors



- Flight Processor
  - glider.app (persistor) or flash-flight.gex (stm32)
  - masterdata
  - autoexec.mi
  - longterm.dat
  - missions
  - ma files

--------Clothesline---------

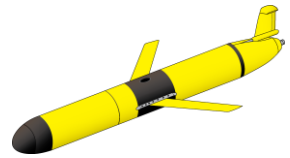
- Science Processor
  - supersci.app (persistor) or flash-science.gex (stm32)
  - proglets.dat

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Glider Variables – *Prefixes*



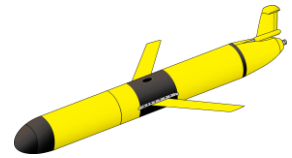
Prefix	Significance
m_	Measured
c_	Commanded
u_	User-defined before run-time
f_	Set in factory
x_	Computed at run time (never set these values!)
s_	Simulated state variables
sci_	Science sensor

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Glider Data Files



	Flight Side	Science Side	*New* Compressed files
ALL data	*.DBD	*.EBD	*.DCD / *.ECD
Small data files (user defined)	*.SBD	*.TBD	*.SCD / *.TCD
Medium data files (user defined)	*.MBD	*.NBD	*.MCD / *.NCD
Log files	*.MLG	*.NLG	*.MCG / *.NCG

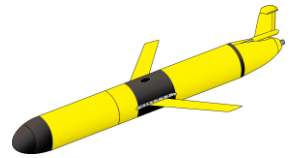
- Logging on/off can create these 8 file types while in gliderdos or lab_mode
- These log files are created at all times when mission is running (except when sending files or receiving files)
- SBD and TBD are customized to send over iridium comms and sent by SFMC script

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Glider Data Files – *Ascii Header*



capx594-2021-195-1-16.sbd

C: > Users > cgoodrich > Downloads > capx594-2021-195-1-16.sbd

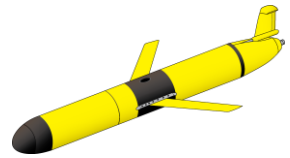
```
dbd_label:              DBD(dinkum_binary_data)file
encoding_ver:           5
num_ascii_tags:         14
all_sensors:            F
the8x3_filename:        03100016
full_filename:          capx594-2021-195-1-16
filename_extension:     sbd
mission_name:           sfmc.mi
fileopen_time:          Thu_Jul_15_17:24:42_2021
total_num_sensors:      2678
sensors_per_cycle:      14
state_bytes_per_cycle:  4
sensor_list_crc:        88a952b1
sensor_list_factored:   1
sa4□y□□B□i~T4o□Ad□□□□□□□□(□*□@□□(\□x□□□□□□□□"□y)□@鶺□wx□□
□□<□A□□@@□+□=□□ A□□□□□-m=d□□□P□@□□w)□@Q□□swx
□swx□□/□□F□□
```


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Common Glider Commands



```
help
lab_mode
wiggle
use
report ++
get
put
! (bang)
consci
whoru
where
```

```
zero_ocean_pressure
dir
run
loadmission
exit
exit reset
```

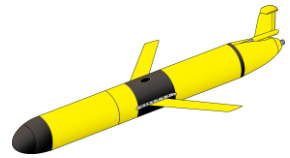
NOTE: List all available commands by typing `help`

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Catching a Glider on Start Up



You should begin these commands when the glider says:

```
SEQUENCE: About to run initial.mi
```

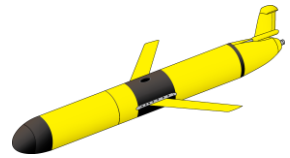
```
> Ctrl-C                  # if successful you should see a GliderDos prompt
> callback 30             # this will hang up the iridium phone for 30 mins.
> lab_mode on             # only if you are in the lab!
> ballast                 # only if you are in the lab!
> report clearall         # to clear buoyancy pump output from 'ballast'
```


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File Transfer Commands – *from the glider*



zs

```
> zs path/fileName
```

- Glider sends file from specified path (from the processor that you are currently talking to)

szs

```
> szs path/fileName
```

- Glider sends file from specified path on the science processor (can only be issued from flight processor)

send

```
> send logFilenames
```

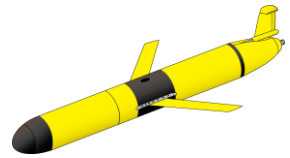
- Glider sends specified log files – path not specified – files are moved to sentlogs once the transfer is complete

s

```
> s logFilenames
```

- Same functionality as ‘send’ but to be used when a glider is in a mission

File Transfer Commands – *to the glider*



zr

```
> zr
```

- Glider receives file - only to be used when not using a dockserver – user selects file that the glider will receive (file lands on processor you are talking to)

dockzr

```
> dockzr fileName
```

- Glider receives file from ‘to-glider’ folder on the dockserver - only to be used with a dockserver (file lands on processor you are talking to)

szr

```
> szr
```

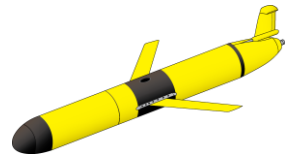
- Science processor receives file - only to be used when not using a dockserver – user selects file that the glider will receive (can only be issued from flight processor)

dockszr

```
> dockszr fileName
```

- Science processor receives file from ‘to-science’ folder on the dockserver - only to be used with a dockserver (can only be issued from flight processor)

Config Directory – *glider configuration files*



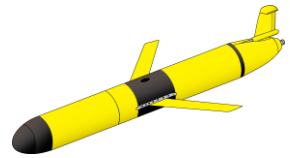
- autoexec.mi
  - Glider's identity file – very important! Not recommended to send over iridium.
- proglets.dat
  - Science identity file (sits on science processor)
- sbdlist.dat
  - Customize content of SBD file - adjust to make .sbd files smaller or larger in size
- tbdlist.dat
  - Customize content of TBD file (sits on science processor) - adjust to make .tbd files smaller or larger in size
- config.srf
  - Customize your surface dialog
- longterm.dat
  - Customize list of sensors that are stored whenever glider powers down

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sbdlist.dat and tbdlist.dat



Control how big the .sbd and .tbd files are so they contain enough real time data to be useful but not so much that the glider spend too much time at the surface transferring over iridium

	Flight Side	Science Side	*New* Compressed files
ALL data	*.DBD	*.EBD	*.DCD / *.ECD
Small data files (user defined)	*.SBD	*.TBD	*.SCD / *.TCD
Medium data files (user defined)	*.MBD	*.NBD	*.MCD / *.NCD
Log files	*.MLG	*.NLG	*.MCG / *.NCG

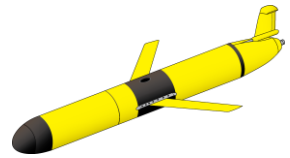


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sbdlist.dat – *defines data in *.sbd files*



Customize content (variables and decimation) :

```
m_present_secs_into_mission
m_present_time
m_water_vx        0        15       -1       -1
m_water_vy
m_de_oil_vol      60       15       -1       -1
c_de_oil_vol      60
m_ballast_pumped  120      15       -1       2
c_ballast_pumped  120      15       -1       2
m_gps_lat  300
m_gps_lon
m_heading      300
m_lat     600
m_lon     600
m_pitch   60       15       -1       2
m_roll    60       15       -1       2
c_wpt_lat
c_wpt_lon
m_battery   3600
m_battpos   30
m_depth     40
```

Must have time variables

Default decimation

If decimation not specified, default will be used

Variable

Interval (sec)

State

Halfyos

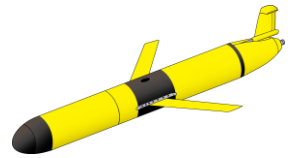
Yo_duty_cycle

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sbdlist.dat – *defines data in *.sbd files*



Customize content (variables and decimation) :

```
m_present_secs_into_mission
m_present_time
m_water_vx          0       15      -1      -1
m_water_vy
m_de_oil_vol        60      15      -1      -1
c_de_oil_vol        60
m_ballast_pumped    120     15      -1      2
c_ballast_pumped    120     15      -1      2
m_gps_lat           300
m_gps_lon
m_heading           300
m_lat               600
m_lon               600
m_pitch             60      15      -1      2
m_roll              60      15      -1      2
c_wpt_lat
c_wpt_lon
m_battery           3600
m_battpos           30
m_depth             40
```

Must have time variables

Default decimation

If decimation not specified, default will be used

Variable

Interval (sec)

State

Halfyos

Yo_duty_cycle

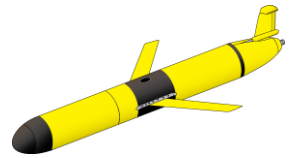
```
# This is a bit-field, combine:
# 8 on_surface, 4 climbing, 2 hovering, 1 diving
# 0  none
# 1  diving
# 2  hovering
# 3  diving|hovering
# 4  climbing
# 5  diving|climbing
# 6  hovering|climbing
# 7  diving|hovering|climbing
# 8  on_surface
# 9  diving|on_surface
# 10 hovering|on_surface
# 11 diving|hovering|on_surface
# 12 climbing|on_surface
# 13 diving|climbing|on_surface
# 14 hovering|climbing|on_surface
# 15 diving|hovering|climbing|on_surface
```


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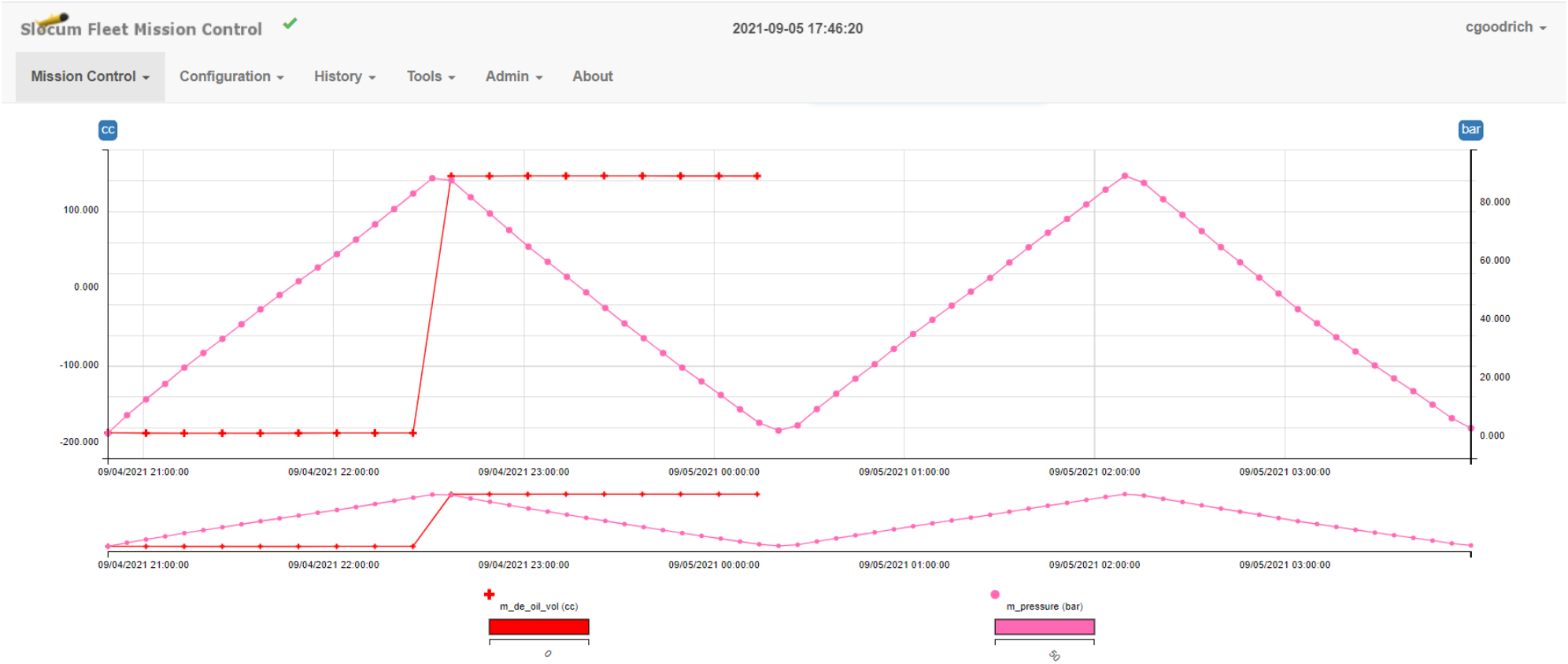
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sbdlist.dat – *defines data in *.sbd files*



Example:

```
m_pressure          360     7
m_de_oil_vol        720     7       2
```

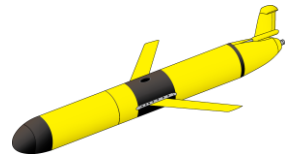


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tbdlist.dat – *defines data in *.tbd files*



Customize content (variables and decimation) :

```
SCI_M_PRESENT_TIME
SCI_M_PRESENT_SECS_INTO_MISSION
SCI_WATER_COND                  30      15      3
SCI_WATER_TEMP                  30      15      3
SCI_WATER_PRESSURE              30      15      3
```

Must have time variables

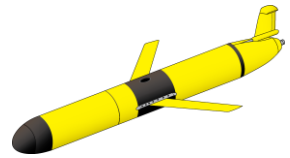
- Same syntax as sbdlist.dat but only for science variables
- Sits on science processor (c:/config/tbdlist.dat)

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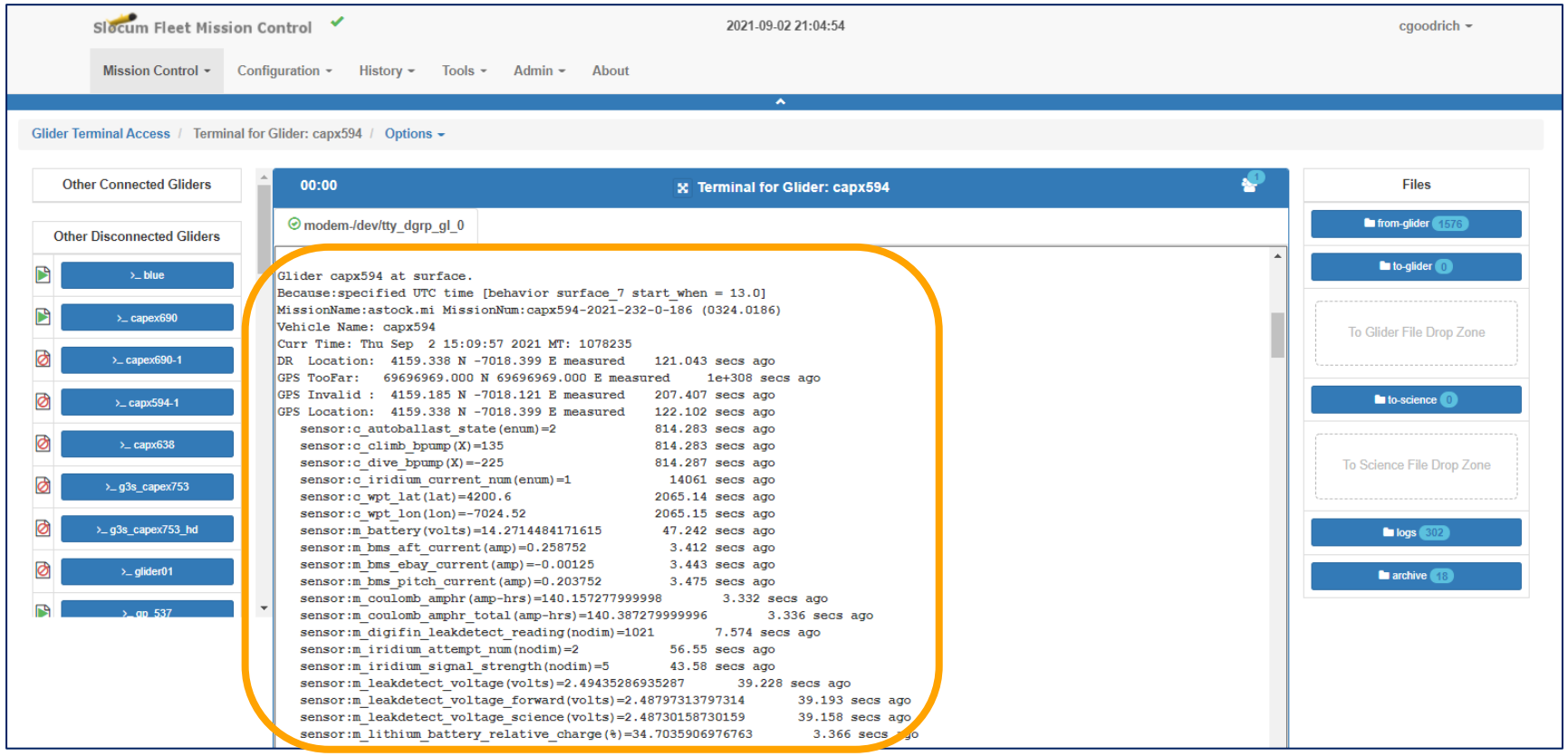
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Surface Dialog – *output when the glider 'surfaces'*

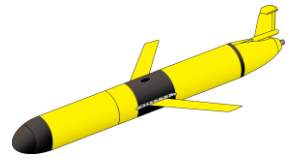


Defined in the file config.srf – can be customized



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longterm.dat – *variables saved when glider powers down*



Oil bladders/bellaframs require replacement every 20,000 half cycles

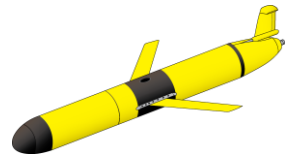
Glider's "fuel gauge"

★ m_avg_climb_rate
m_avg_upward_inflection_time
★ m_tot_num_inflections
m_tot_horz_dist
m_lat
m_lon
m_tot_ballast_pumped_energy
m_battery
m_iridium_call_num
m_iridium_dialed_num
★ m_coulomb_amphr_total
s_water_depth_avg
s_water_depth_delta
s_water_depth_wavelength
f_ocean_pressure_min
m_avg_speed
x_last_wpt_lat
x_last_wpt_lon
x_de_avg_oil_vol_ierr_on_ascent
x_de_avg_oil_vol_ierr_on_descent
x_hover_ballast_shallow
x_hover_ballast_deep
x_hover_depth_shallow
x_hover_depth_deep

- ***Customize list of sensors that are stored whenever glider powers down***

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Installed Devices



- Devices are defined in at the beginning of autoexec.mi
- ‘use’ command or ‘Ctrl-W’ in a mission

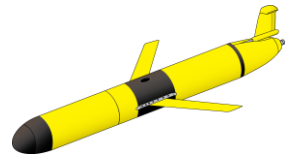
```
                name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                     [I Installed] [- Not_Installed]
                       [u In_use] [- Not_In_use] [X Out_of_Service]
                name          limits      stats (#total/#mission/#segment)
 0          simdrvr  -
 1      test_driver  -
 2            ARGOS* I u  -1  20   5  0
 3         WATCHDOG  I u  -1  -1  -1  0
 4          DEADMAN  I u  -1  20   5  0
 5          CONSOLE* I u  -1  20   5  0
 6              GPS  I u  -1  20   5  0
 7           pinger  -
 8         attitude  -
 9   attitude_tcm3  -
10    attitude_rev  I u   3  20   5  0 [   0   0  0] [   1   0  0] [   0   0  0]
11  ocean_pressure  I u   3  20   5  0
12          vacuum  I u   3  20   5  0
13         battery  I u   3  20   5  0
14 lithium_battery  -
15         air_pump  I u   3  20   5  0
16      pitch_motor  I u   3  20   5  0
17    science_super  I u   3  20   5  0
18       roll_motor  -
19      fpitch_pump  -
20        fin_motor  -
21          digifin  I u   3  20   5  0
22        altimeter  I u   3  20   5  0
23              ctd  -
24          IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   0   0  0] [   2   0  0]
25       leakdetect  I u   3  20   5  0
26         recovery  I u   3  20   5  0
27          coulomb  I u   3  20   5  0
28         veh_temp  I u   3  20   5  0
29    BUOYANCY_PUMP  I u   3  20   3  0
30 THERMAL_ACC_PRE  -
31  THERMAL_ENGINE  -
32    THERMAL_PUMP  -
33          DE_PUMP  -
34      thruster_g1  -
35         thruster  I u   3  20   5  0
devices:(t/m/s) errs:   0/   0/   0 warn:   1/   0/   0 odd:   2/   0/   0
```


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Installed Devices



```
                name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                     [I Installed] [- Not_Installed]
                       [u In_use] [- Not_In_use] [X Out_of_Service]
                name          limits     stats (#total/#mission/#segment)
                              os w/s w/m os       errors      warnings      oddities
 0          simdrvr  I u    3  20   5  0
 1      test_driver  I u    3  20   5  0
 2           ARGOS* I u   -1  20   5  0
 3         WATCHDOG  I u   -1  -1  -1  0
 4         DEADMAN  I u   -1  20   5  0
 5         CONSOLE* I u   -1  20   5  0
 6             GPS  I u   -1  20   5  0
 7     attitude_rev  I u    3  20   5  0
 8  ocean_pressure  I u    3  20   5  0
 9          vacuum  I u    3  20   5  0
10         battery  I u    3  20   5  0
11        air_pump  I u    3  20   5  0
12      pitch_motor  I u    3  20   5  0
13    science_super  I u    3  20   5  0
14         digifin  I u    3  20   5  0
15       altimeter  I u    3  20   5  0
16    altimeter_232  -
17         IRIDIUM* I u   -1  20   5  0 [   0   0  0] [   0   0  0] [   2   2  2]
18      leakdetect  I u    3  20   5  0
19        recovery  -
20         coulomb  I u    3  20   5  0
21        veh_temp  I u    3  20   5  0
22    BUOYANCY_PUMP  I u    3  20   3  0
23         DE_PUMP  -
24         HD_PUMP  -
25         thruster  I u    3  20   5  0
devices:(t/m/s) errs:   0/   0/   0 warn:   0/   0/   0 odd:   2/   2/   2
```

Type `use - sensor_name` to temporarily remove devices

Type `use + sensor_name` to reinstall devices that have been taken out of service

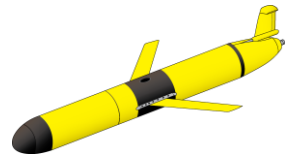
Type `use all` or `use none` to install or remove all devices

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Installed Devices



```
                name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                     [I Installed] [- Not_Installed]
                       [u In_use] [- Not_In_use] [X Out_of_Service]
                name          limits      stats (#total/#mission/#segment)
 0          simdrvr  -
 1      test_driver  -
 2           ARGOS* I u  -1  20   5  0
 3         WATCHDOG  I u  -1  -1  -1  0
 4          DEADMAN  I u  -1  20   5  0
 5          CONSOLE* I u  -1  20   5  0
 6              GPS  I u  -1  20   5  0
 7           pinger  -
 8         attitude  -
 9    attitude_tcm3  -
10     attitude_rev  I u   3  20   5  0 [   0   0  0] [   1   0  0] [   0   0  0]
11  ocean_pressure  I u   3  20   5  0
12          vacuum  I u   3  20   5  0
13         battery  I u   3  20   5  0
14 lithium_battery  -
15         air_pump  I u   3  20   5  0
16      pitch_motor  I u   3  20   5  0
17    science_super  I u   3  20   5  0
18       roll_motor  -
19      fpitch_pump  -
20        fin_motor  -
21          digifin  I u   3  20   5  0
22        altimeter  I u   3  20   5  0
23              ctd  -
24          IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   0   0  0] [   2   0  0]
25       leakdetect  I u   3  20   5  0
26         recovery  I u   3  20   5  0
27          coulomb  I u   3  20   5  0
28         veh_temp  I u   3  20   5  0
29    BUOYANCY_PUMP  I u   3  20   3  0
30 THERMAL_ACC_PRE  -
31  THERMAL_ENGINE  -
32     THERMAL_PUMP  -
33          DE_PUMP  -
34      thruster_g1  -
35          thruster  I u   3  20   5  0
devices:(t/m/s) errs:   0/   0/   0 warn:   1/   0/   0 odd:   2/   0/   0
```

Total # of errors

of errors in segment

of errors in mission

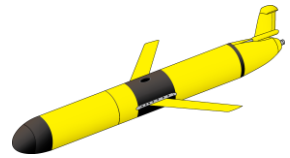
'u' indicates in use

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Installed Devices



```
                name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                     [I Installed] [- Not_Installed]
                       [u In_use] [- Not_In_use] [X Out_of_Service]
                name          limits      stats (#total/#mission/#segment)
 0          simdrvr  -
 1      test_driver  -
 2            ARGOS* I u  -1  20   5  0
 3         WATCHDOG  I u  -1  -1  -1  0
 4          DEADMAN  I u  -1  20   5  0
 5          CONSOLE* I u  -1  20   5  0
 6              GPS  I u  -1  20   5  0
 7           pinger  -
 8         attitude  -
 9   attitude_tcm3  -
10    attitude_rev  I u   3  20   5  0 [   0   0  0] [   1   0  0] [   0   0  0]
11 ocean_pressure  I u   3  20   5  0
12          vacuum  I u   3  20   5  0
13         battery  I u   3  20   5  0
14lithium_battery  -
15         air_pump  I u   3  20   5  0
16     pitch_motor  I u   3  20   5  0
17   science_super  I u   3  20   5  0
18      roll_motor  -
19     fpitch_pump  -
20       fin_motor  -
21         digifin  I u   3  20   5  0
22       altimeter  I u   3  20   5  0
23             ctd  -
24          IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   0   0  0] [   2   0  0]
25      leakdetect  I u   3  20   5  0
26        recovery  I u   3  20   5  0
27         coulomb  I u   3  20   5  0
28        veh_temp  I u   3  20   5  0
29   BUOYANCY_PUMP  I u   3  20   3  0
30THERMAL_ACC_PRE  -
31  THERMAL_ENGINE  -
32    THERMAL_PUMP  -
33          DE_PUMP  -
34     thruster_g1  -
35         thruster  I u   3  20   5  0
devices:(t/m/s) errs:   0/   0/   0 warn:   1/   0/   0 odd:   2/   0/   0
```

Total # of warnings

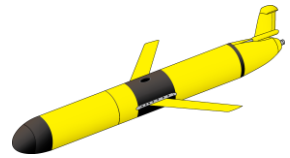
of warnings in segment

of warnings in mission

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Installed Devices



```
                name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
                     [I Installed] [- Not_Installed]
                       [u In_use] [- Not_In_use] [X Out_of_Service]
                name          limits      stats (#total/#mission/#segment)
 0          simdrvr  -
 1      test_driver  -
 2            ARGOS* I u  -1  20   5  0
 3         WATCHDOG  I u  -1  -1  -1  0
 4          DEADMAN  I u  -1  20   5  0
 5          CONSOLE* I u  -1  20   5  0
 6              GPS  I u  -1  20   5  0
 7           pinger  -
 8         attitude  -
 9    attitude_tcm3  -
10     attitude_rev  I u   3  20   5  0 [   0   0  0] [   1   0  0] [   0   0  0]
11  ocean_pressure  I u   3  20   5  0
12           vacuum  I u   3  20   5  0
13          battery  I u   3  20   5  0
14lithium_battery  -
15         air_pump  I u   3  20   5  0
16      pitch_motor  I u   3  20   5  0
17    science_super  I u   3  20   5  0
18       roll_motor  -
19      fpitch_pump  -
20        fin_motor  -
21          digifin  I u   3  20   5  0
22        altimeter  I u   3  20   5  0
23              ctd  -
24          IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   0   0  0] [   2   0  0]
25       leakdetect  I u   3  20   5  0
26         recovery  I u   3  20   5  0
27          coulomb  I u   3  20   5  0
28         veh_temp  I u   3  20   5  0
29    BUOYANCY_PUMP  I u   3  20   3  0
30THERMAL_ACC_PRE  -
31  THERMAL_ENGINE  -
32     THERMAL_PUMP  -
33          DE_PUMP  -
34      thruster_g1  -
35          thruster  I u   3  20   5  0
devices:(t/m/s) errs:   0/   0/   0 warn:   1/   0/   0 odd:   2/   0/   0
```

Total # of oddities

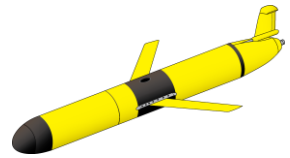
of oddities in segment

of oddities in mission

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Mission Planning



Ballast & H-Moment

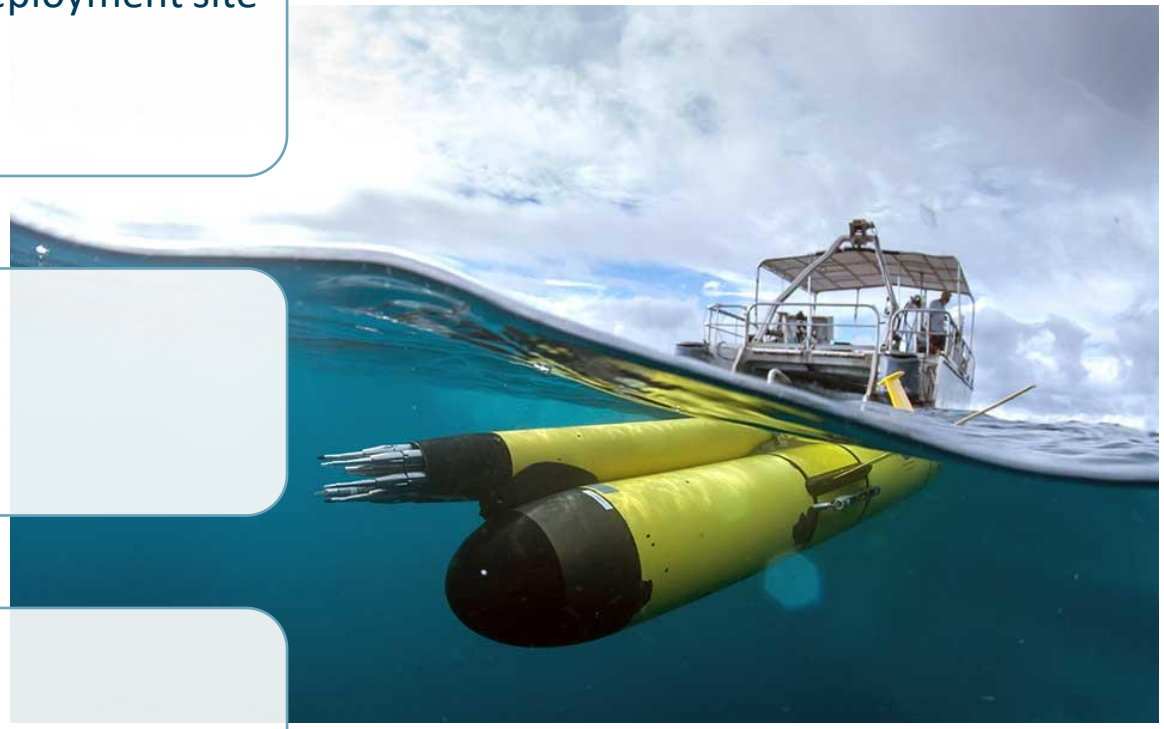
- Identify surface Temp & Salinity conditions at deployment site
- 5 < H-moment < 6 (ideal)
- Final Seal

Testing & Functional

- Vacuum test
- Functional checkout procedure

Simulate Missions

- Easy: modify existing missions
- Harder & riskier without experience: write your own

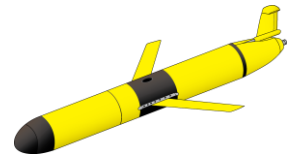


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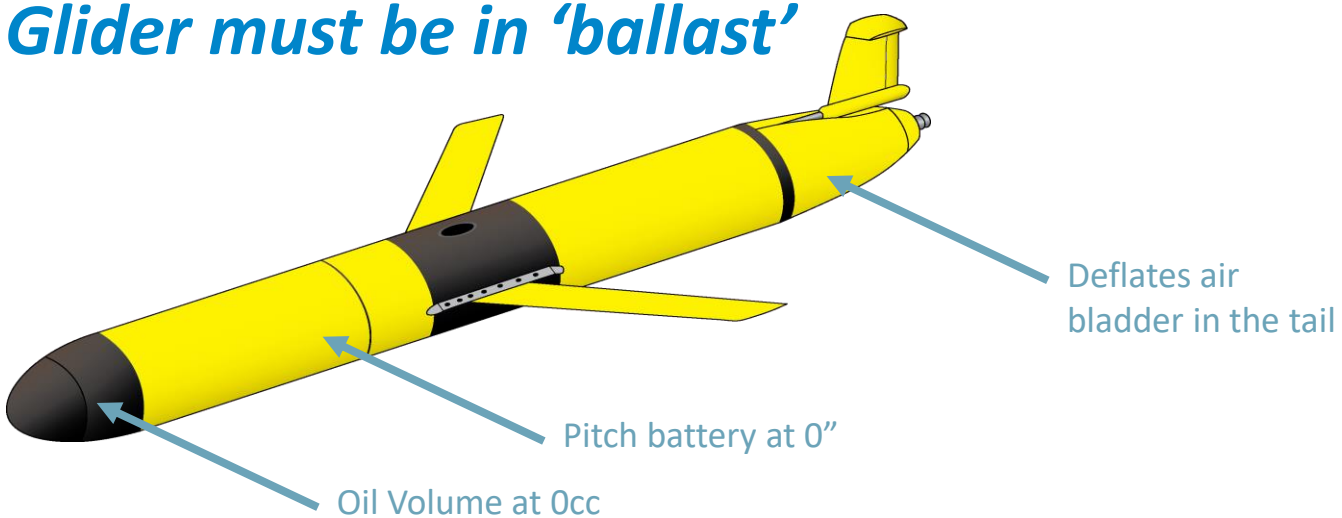
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Ballasting



Glider must be in 'ballast'



Goal is to have the glider neutrally buoyant for the surface of the target water.

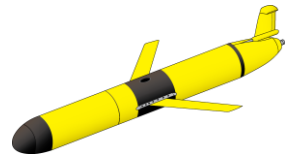
Need to know Temp & Salinity/Cond/Density of where the glider is being deployed

ρ = density
m = mass
V = volume

$$\rho = \frac{m}{V}$$

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Mission Commands – *how to start a glider mission*



loadmission

```
> loadmission mission1.mi
```

- Sets glider sensor values found in mission1.mi (does not start an actual mission)

run

```
> run mission2.mi
```

- Instructs glider to begin running mission2.mi

sequence

```
> sequence mission3.mi mission4.mi mission5.mi
```

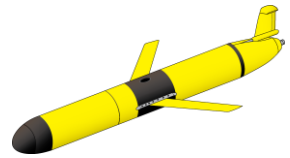
- Instructs glider to run mission3.mi. When mission3.mi completes glider will run mission4.mi, and when that finishes it will run mission5.mi

OR

```
> Sequence mission5.mi(5)
```

- Instructs glider to run mission5.mi five times in a row

Testing Your New Glider



- Make sure to confirm the following:
  - FreeWave comms
  - Iridium comms
  - Wiggle (exercise all motors)
  - Science sensor output

Functional Checkout Procedure (4095-FCP)

- Personalize your glider!
  - Rename it
  - Add your phone numbers
  - Connect to your dockserver

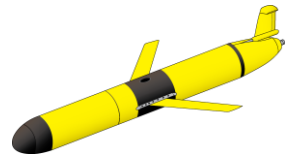


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Glider Dos & Don'ts



DO

- Secure it properly in crate with all three straps for shipping.
- Use fresh desiccants on each deployment.
- Monitor internal vacuum before launch (less vacuum indicates a leak; positive pressure may indicate dangerous gas accumulation).
- Simulate missions before launch.
- Test Iridium and Argos telemetry before launch.
- Burn off passivation in lithium batteries with a ~500m dive

DON'T

- Never power up a shallow glider without a vacuum.
- Never run a simulation on a glider other than "on_bench."
- Never deploy a glider in simulation.
- Never deploy a glider in "boot pico"(persistor) or "boot shell"(stm32)
- Never exit to pico (persistor) or exit to shell (stm32) during a deployment.
- Never power on a glider with more than 15 vDC from an external power supply.
- Never deploy a glider in lab_mode.
- Never perform the top of a yo below 30 meters (with 100 or 200 meter glider).
- Never secure the glider to the cart while over railing or in the water.

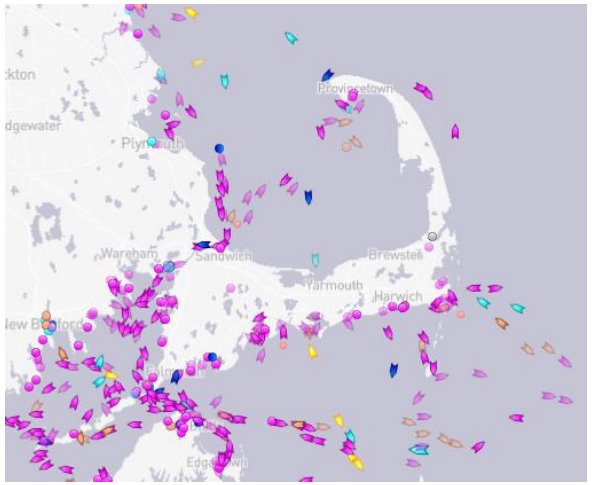
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Water Space Management









Consider the following when planning your deployment:

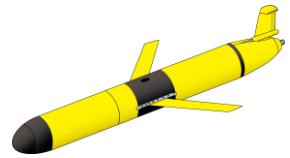
- Tides
- Shipping lanes
- Recreational traffic
- Fishing grounds
- Bathymetry & sea floor characteristics

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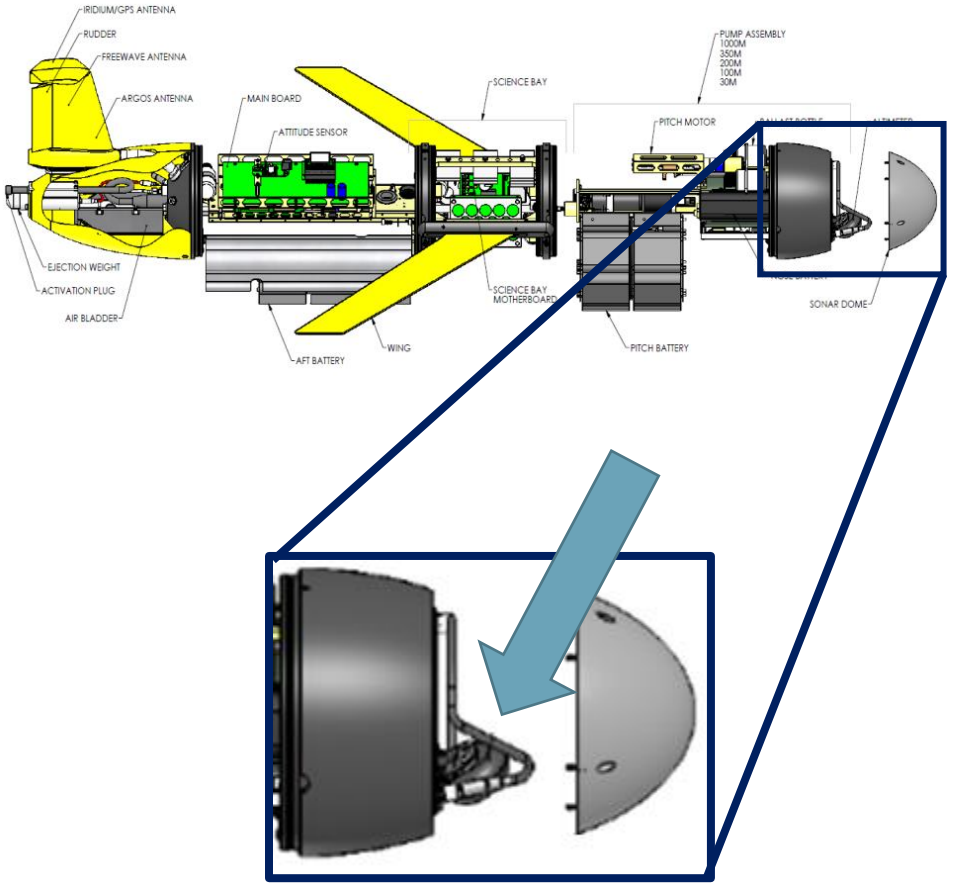
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Altimeters



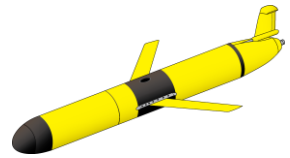
- The altimeter is downward looking
- Avoid flying close to bottom in dynamic terrain
- Altimeters use energy – Pilots might choose to turn off when not needed:
- sensor: u_alt_min_depth(m) 2.0 #how deep vehicle must be to use altitude



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Battery Options



Alkaline Batteries

- Most Affordable Option
- Least Endurance
- Lightest (easy to ballast)
- Number of batteries may vary depending on payload

Lithium Primary Batteries

- Highest Endurance
- 3s (12V) and 4s (15V) options
- Requires Enable Circuit
- Proper shipping certification needed for Hazardous goods

Lithium Rechargeable Batteries

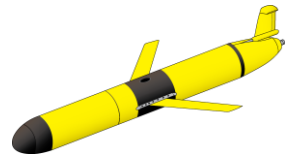
- Rechargeable Battery Option
- Requires Battery Management System Board
- Proper shipping certification needed for Hazardous goods

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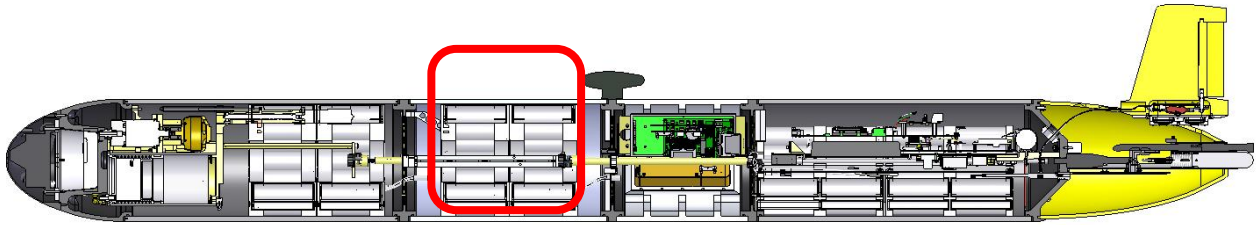
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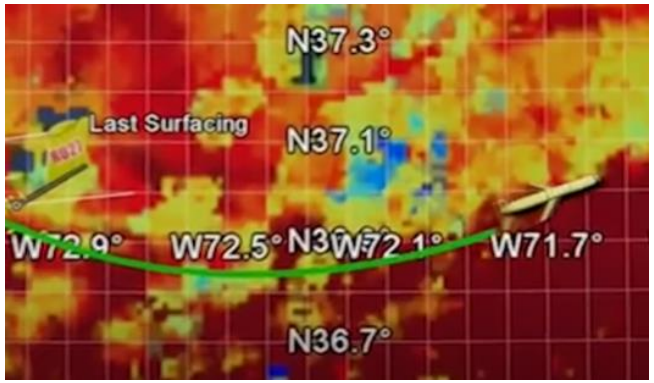
Glider Endurance

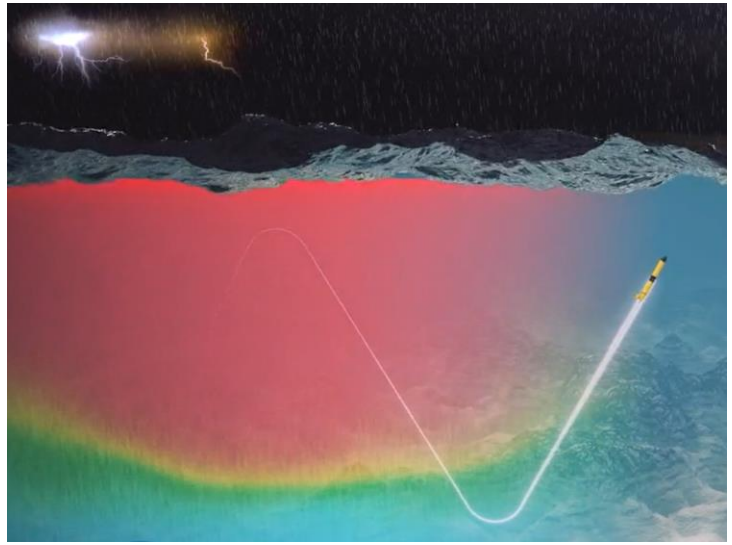


What factors come into play?

- Glider Build
  - Which batteries used?
  - Extended endurance payload?
  - Sensors installed
- Mission length
  - Power consumption
- Sampling strategy
  - How frequently do you need science data?
- Do you need to use the altimeter?
  - Can be uninstalled in deep water
  - Remember not to overwrite altimeter settings in mission





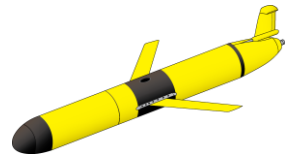


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Coulomb Counter and Battery Capacities



- The coulomb counter is the glider's "fuel tank"
- Monitor both amp-hrs and voltage through the deployment!
- Important for lithium battery sets.
  - Gliders will begin aborting for low remaining energy when `m_coulomb_amphr_total` reaches 10% remaining)
  - zero this value when batteries are replaced:

    ```
    put m_coulomb_amphr_total 0
    ```
- User must set f_coulomb_battery_capacity for each battery type (typically done in autoexec.mi)

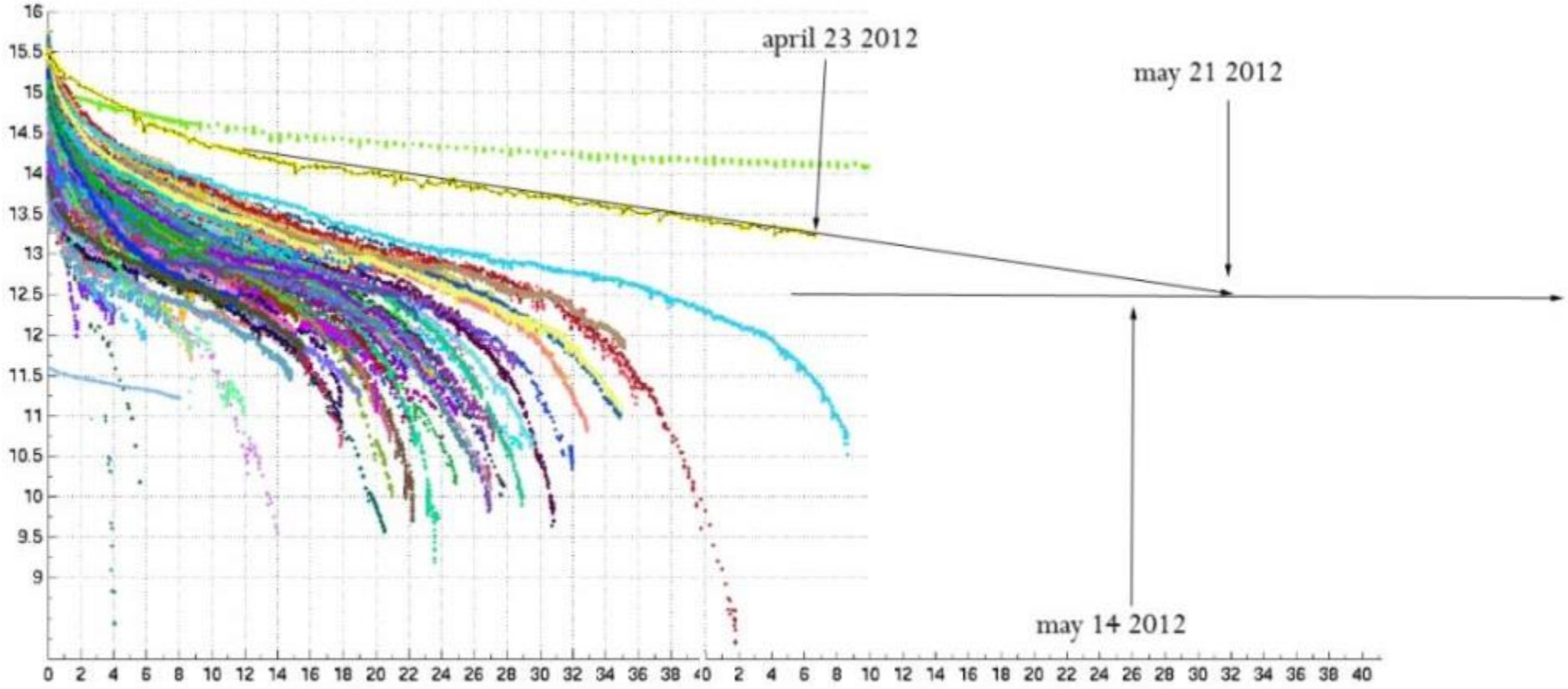
	Total Non-derated Amphrs	Total Non-derated Watt Hours	Recommended f_coulomb_battery_capacity(amp-hrs)	Recommended Undervolts(V)
Alkaline Standard	168	2520	120	10
3s Primary Standard	780	8424	720	9.8
3s Primary Extended	1140	12312	1050	9.8
4s Primary Standard	600	8600	550	12
4s Primary Extended	870	12500	800	12
Lithium Rechargeable Standard	234.6	3300	215	12.5
Lithium Rechargeable Extended	326.4	4700	300	12.5

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Voltage Curve – *alkaline*

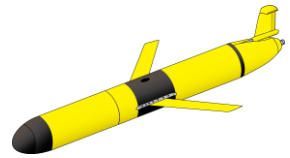


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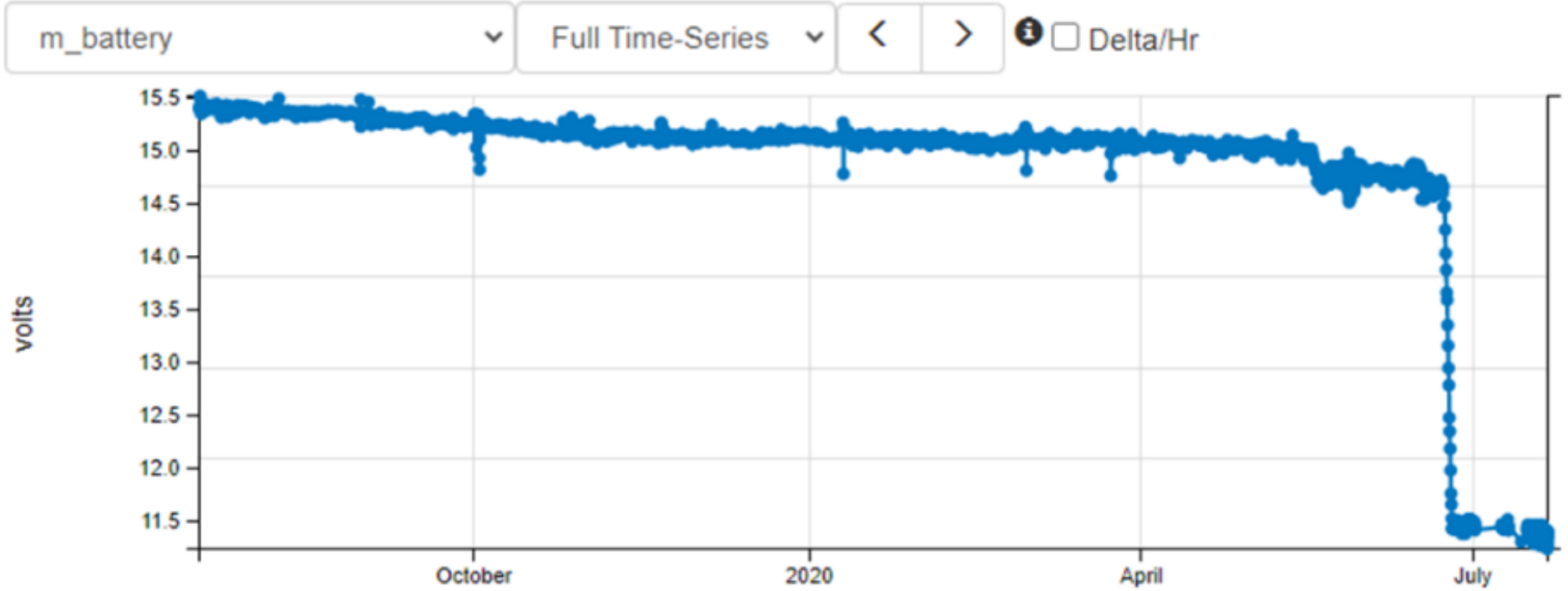
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Voltage Curve – *4S lithium batteries*



Example of 4S voltage curve

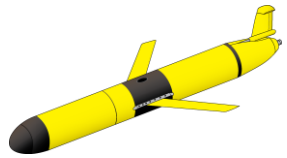
(this glider was on a low power mission with energy bay)

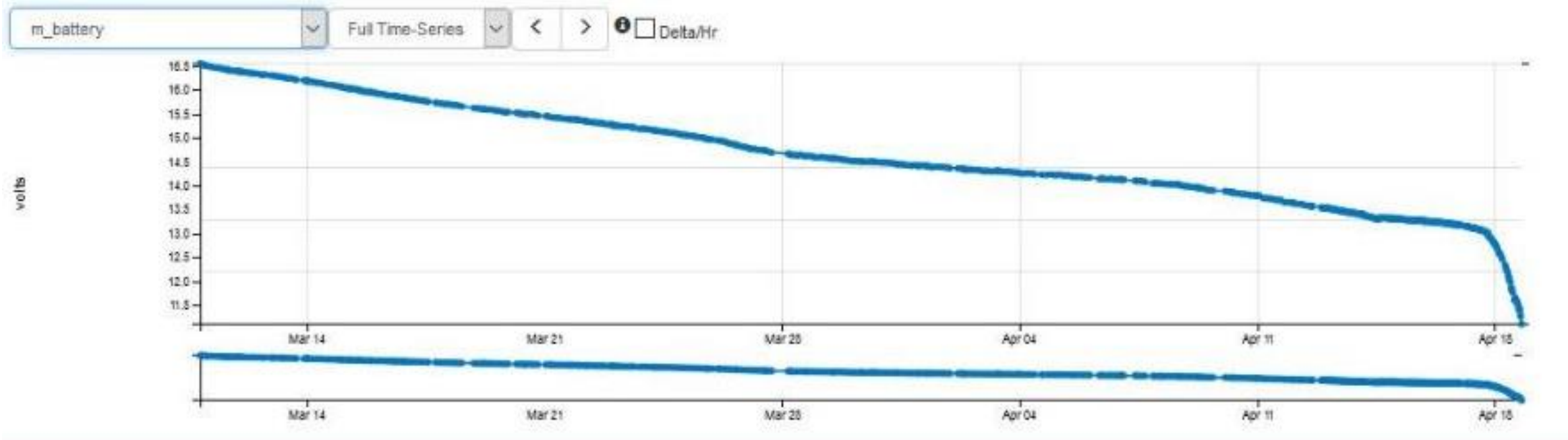


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Voltage Curve – *Rechargeable lithium batteries*





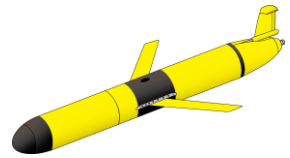
221 total coulomb amphr when they shut off a 10v

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Crisis and Mission End Management



Safely getting your Glider home

Make sure you have answers to the following questions when it comes time to recover your glider:

- Who?
  - Which members of your team are responsible and available to recover the glider?
  - Who is backup? Who can help?
- Where?
  - Is your glider in an easy-to-recover area?
  - How will you get to/from this area?
- When?
  - When will the recovery occur?
  - What is your plan if the glider aborts early?
- How?
  - What resources / ships / boats etc. are needed to recover the glider?

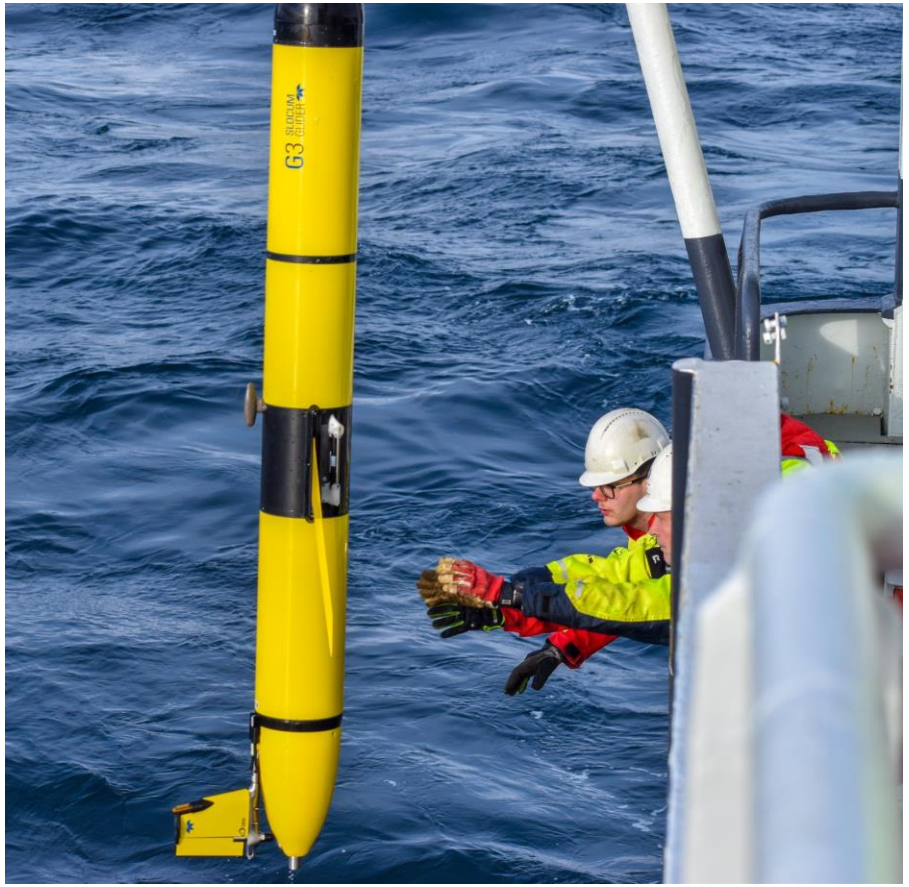


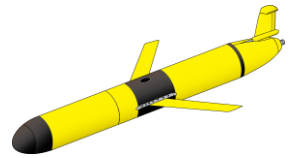
Photo: Fiona Elliot

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The Glider Aborted – *now what?*



- **Send the following commands to the glider:**
  - `why?` (explains why glider aborted)
  - `use` (prints list of installed devices)
  - `where` (prints surface dialog to screen) (callback 2 0?)
- **Download the relevant MLG (maybe the DBD)**
- **Increase time in GliderDOS so that glider doesn't sequence into `lastgasp.mi`**
- **Run callback 30 script**
  - Remember: GPS & Iridium share the same antenna. Only one works at a time!



https://datahost.webbresearch.com/files.php?cwd=/glider/production/doco/how-it-works/abort-sequuences.txt

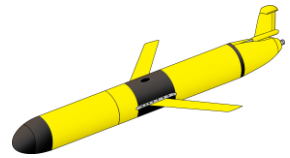
https://datahost.webbresearch.com/viewforum.php?f=7

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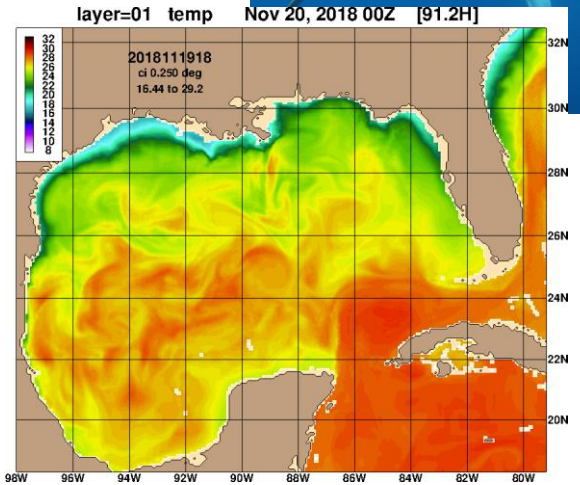
Dangers to Gliders













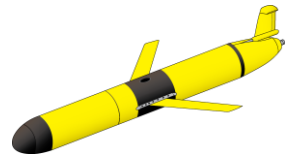
- People – strangers and technicians
- Shipping (transporting)
- Predators (Remus Great White Shark video)
- Seals (green plug)
- Remora
- Low density surface layer (warm and fresh H20)
- Trucks

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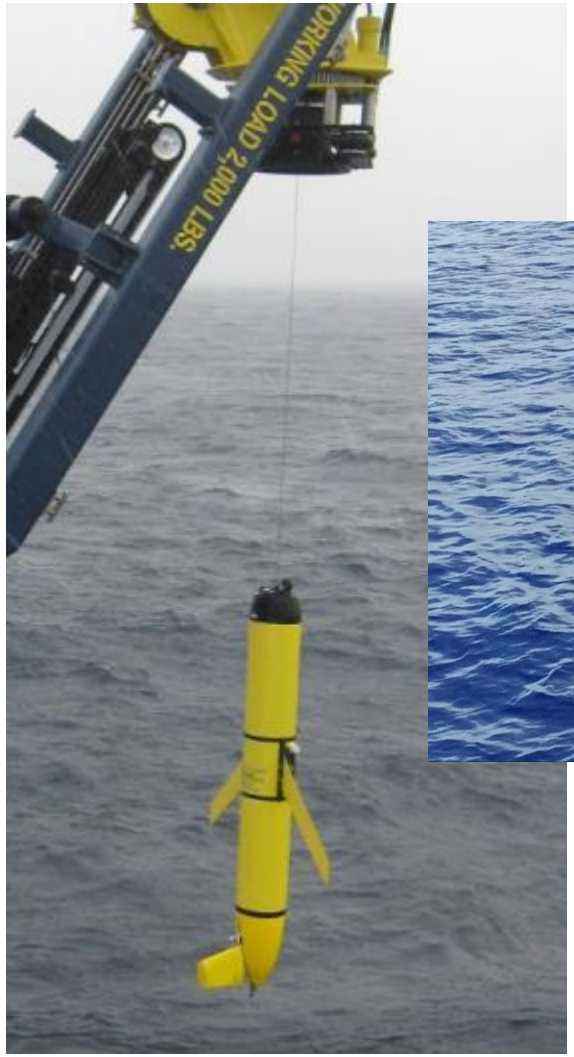
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Recovery



- Small boat?
- Big boat?
- Strobe on?
- Recovery system?
- GPS format on recovery boat: is translation needed?
- ARGOS data?

MAKE A PLAN!



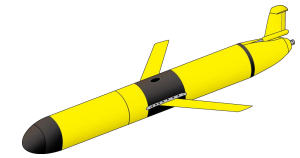


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Mission Writing

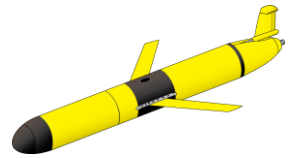




- **Start simple!**
- **Modify astock.mi**
  - Default mission
  - flies around 4 points in Ashumet Pond – where Teledyne Webb Research regularly flies and tests gliders
- **You will likely need to adjust:**
  - d_target_depth
  - max_wpt_distance
  - Surface intervals
    - No comms timeout
    - Every x minutes
    - Surfacing at every waypoint

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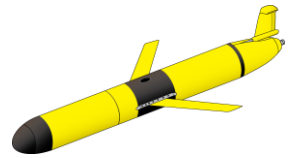
astock.mi



- **astock.mi Behaviors:**
  - abend (abort behaviors)
  - surface (surface behavior, surfac21.ma)
  - surface (surface behavior, surfac22.ma)
  - surface (surface behavior, surfac23.ma)
  - surface (surface behavior, surfac24.ma)
  - surface (surface behavior, surfac25.ma)
  - surface (surface behavior, surfac26.ma)
  - goto (waypoints, goto_l10.ma)
  - yo (dive and climb behavior, yo14.ma)
  - sample (science control, sample10.ma)
  - prepare_to_dive (needs to be here, but don’t change)
  - sensors_in (needs to be here, but don’t change)

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Mission Writing



Modify Waypoints

- The goto_l10.ma file controls waypoints

```
<start:b_arg>
b_arg: num_legs_to_run(nodim) -1 # loop
b_arg: start_when(enum) 0 # BAW_IMMEDIATELY
b_arg: list_stop_when(enum) 7 # BAW_WHEN_WPT_DIST
b_arg: initial_wpt(enum) -2 # closest
b_arg: num_waypoints(nodim) 4
<end:b_arg>
<start:waypoints>
-7032.0640   4138.1060
-7031.9200   4138.1090
-7031.9170   4138.0000
-7032.0610   4137.9980
<end:waypoints>
```

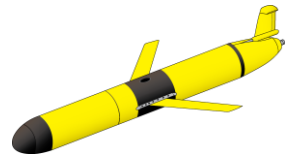


Modify Dive Characteristics

- The yo14.ma file controls:
  - Dive to depth
    - must be less than max working depth
  - Climb to depth
    - must be shallower than 30m for piston-driven gliders
    - Inflecting at depths deeper than 30m will cause mission to abort (preventing damage to hardware)
  - Altitude
    - 6m off bottom for piston-driven gliders
    - 15m off bottom for oil-driven gliders

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Mission Writing

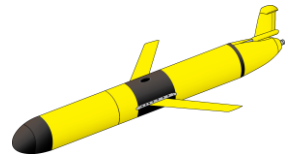


Surface Behaviors – reasons to come to the surface

- No Comms
- Pitch Idle (pitch not commanded)
- Stack Idle (nothing commanded)
- Heading Idle
- Hit a Waypoint
- Every X seconds
- For UTC time
- When requested by science (most science sensors do not have this control)

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Science Sensors

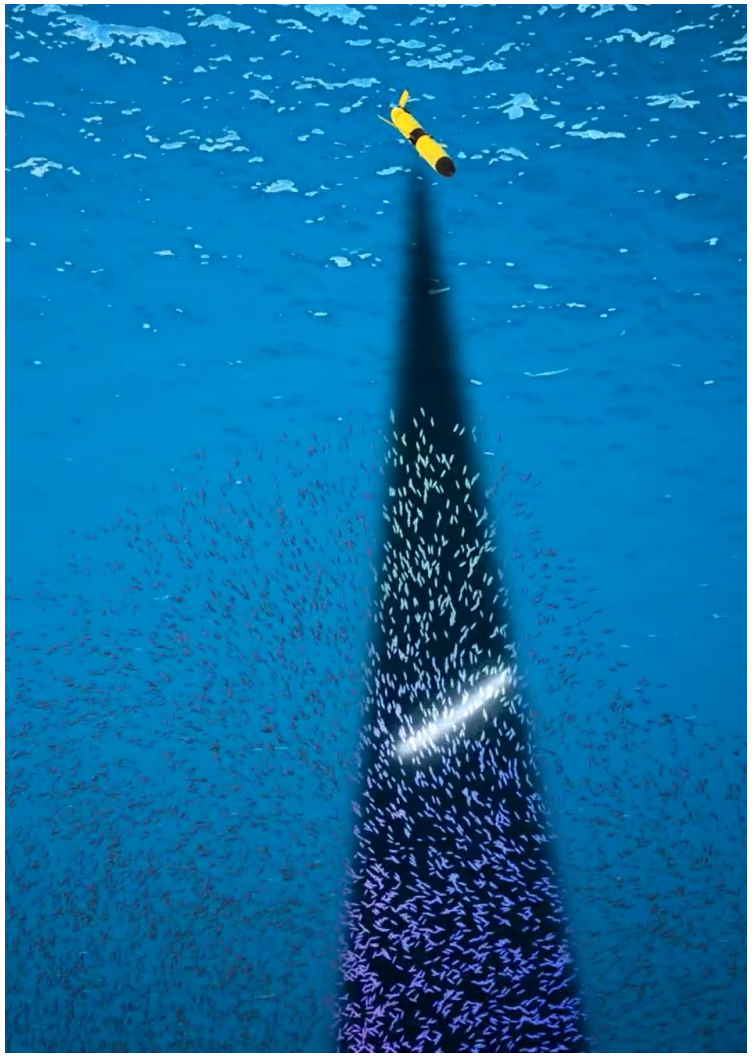


The “clothesline”

- Communication between the Flight & Science processors

Specify sampling scheme in sampleXX.ma files

- state_to_sample (diving, climbing, hovering, on sfc)
- intersample_time (# seconds between measurements)
- nth_yo_to_sample (sample only n^{th} yo after 1st yo)
- intersample_depth (# meters between measurements)
- min_depth
- max_depth

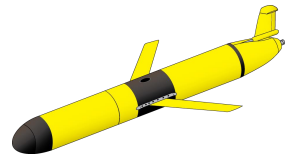


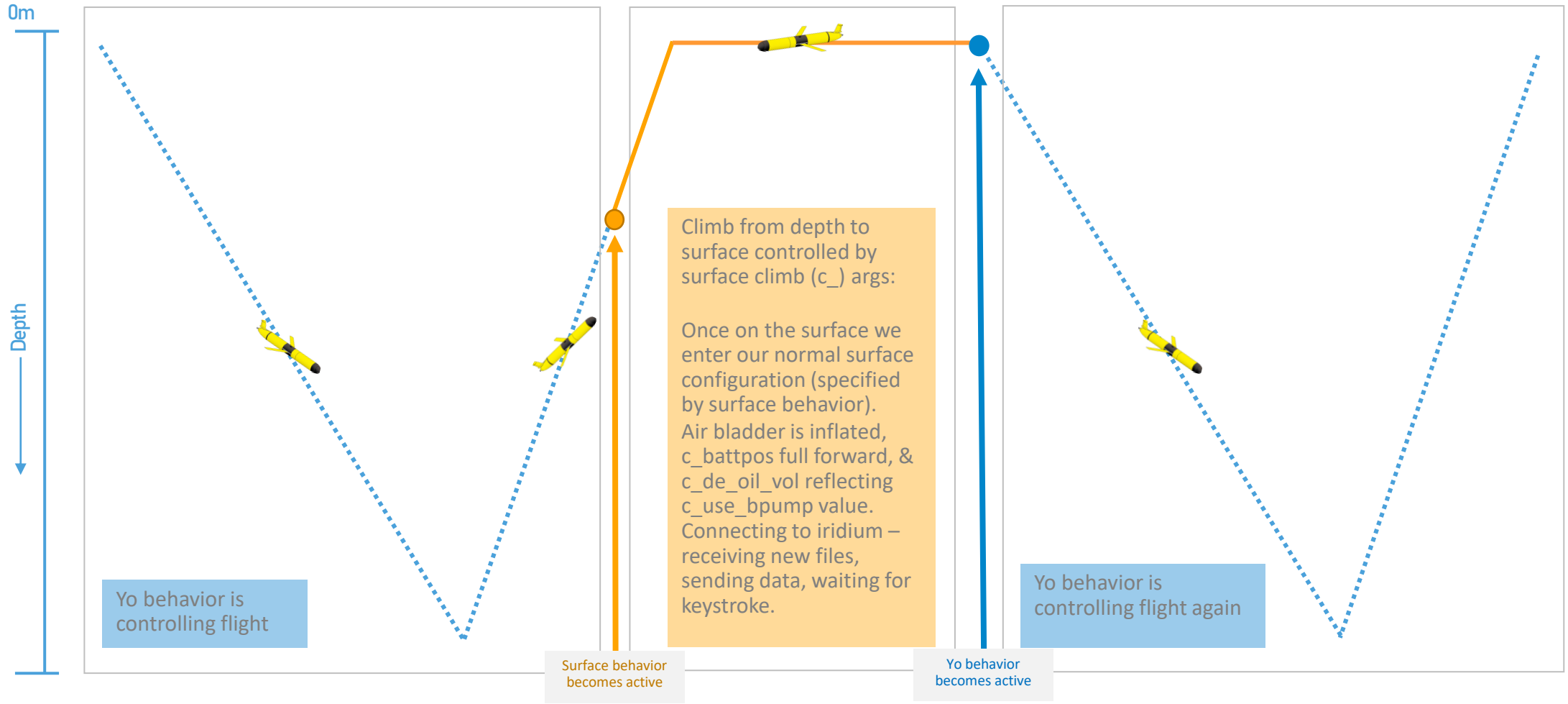
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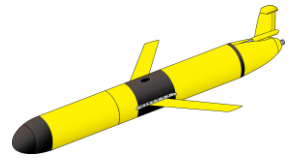
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astock.mi

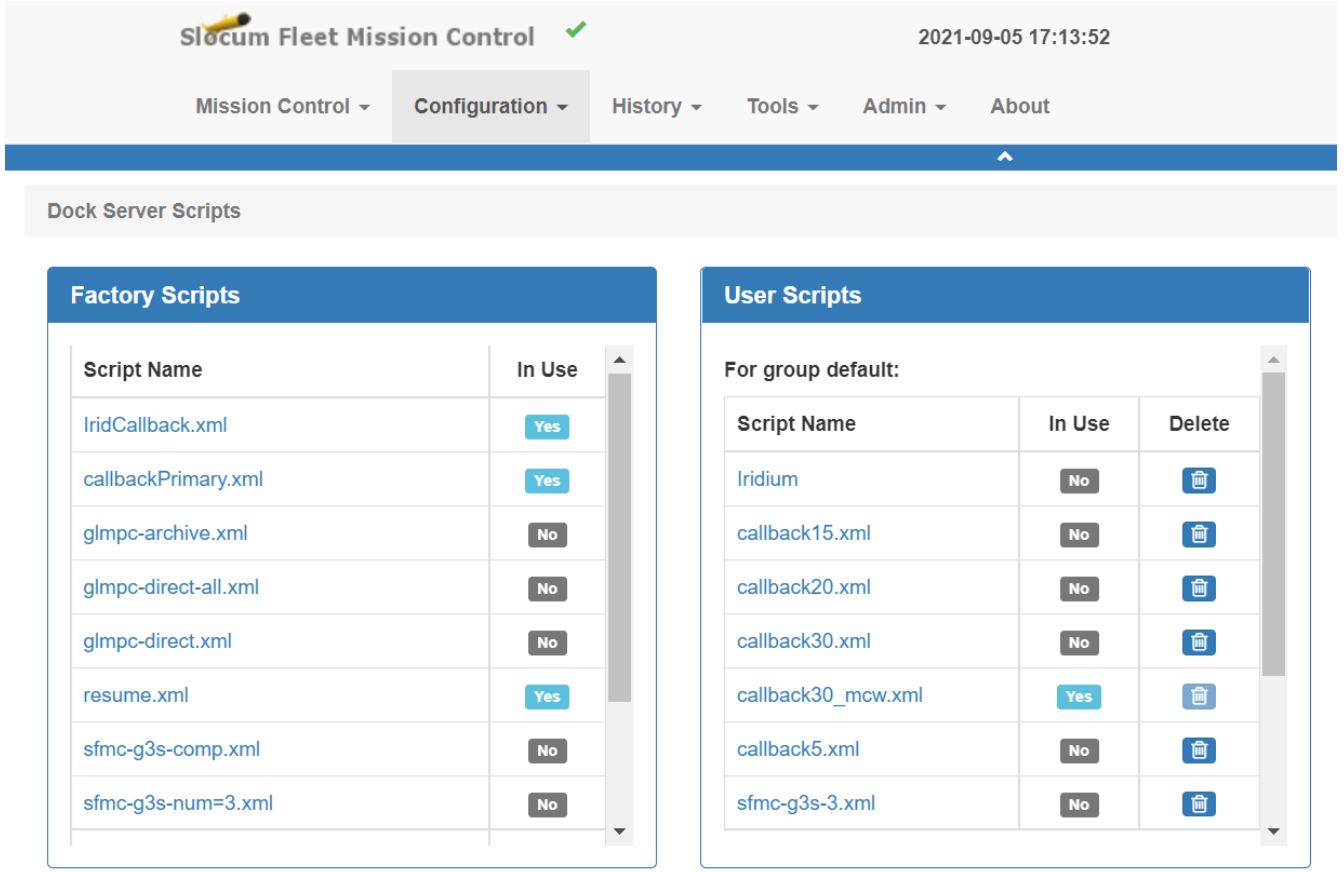



0m
Depth
Yo behavior is controlling flight
Climb from depth to surface controlled by surface climb (c_) args:
Once on the surface we enter our normal surface configuration (specified by surface behavior).
Air bladder is inflated, c_battpos full forward, & c_de_oil_vol reflecting c_use_bpump value.
Connecting to iridium – receiving new files, sending data, waiting for keystroke.
Surface behavior becomes active
Yo behavior becomes active
Yo behavior is controlling flight again

Scripts – *glider autopilot*



- SFMC is TWR recommended
- sends all *.sbd and all *.tbd
  - S *.sbd *.tbd –f irid –num=3
- dockzr all *.ma files
  - !dockzr *.* -f irid
- Ctrl-F – ReRead .ma files
- Ctrl-W – Warning report (prints device list)
- Ctrl-R (dive!)

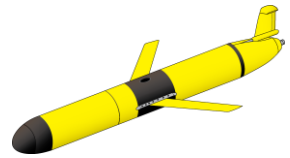


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DBD format – *process glider data files*



- File naming convention
  - File name on glider, example :



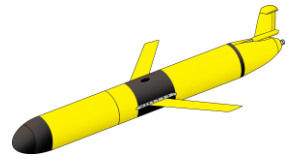
  - Full file name, example :



- Cache files
  - Needed to process data files
- Processing
  - SFMC (processed automatically)
  - Executables (windoze-bin or linux-bin)

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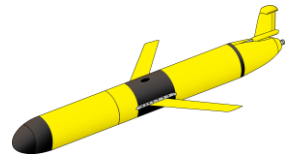
Simulations – *practice before deployment*



- **Three different types of simulator are available**
  - Pocket Simulator (just a persistor or STM32 processor, all electronics and motors are simulated)
  - Shoebox Simulator (persistor or STMS32 processor & mainboard, all motors are simulated; can be connected to a science board)
  - Full glider (all motors & electronics are used, some sensors simulated)
- **simul.sim file controls the type of simulation**
- **This file must be placed in the flight config directory with the appropriate text:**
  - no_electronics (pocket simulator)
  - just_electronics (shoebox simulator)
  - on_bench (full glider)

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Simulations – *putting the glider into simulation mode*



```
simul?                      # verify the glider is not currently in simulation
```

To create the simulation file:

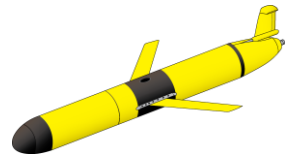
```
capture config/simul.sim
on_bench
ctrl-c
dir config                  # verify simul.sim file is in config directory
Type config/simul.sim       # verify the contents of the simulation file are correct
Exit reset                  # to force the glider to read the simulation file
Simul?                      # verify the glider is in simulation mode
```


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Simulations – *taking the glider out of simulation mode*



```
simul?                  # verify the glider is currently in simulation
```

Remove the simulation file:

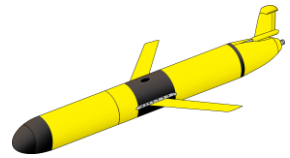
```
del config/simul.sim
dir config              # verify simul.sim file is deleted from config directory
Exit reset              # to force the glider to read the simulation file
Simul?                  # verify the glider is NOT in simulation mode
```


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Simulations – *loading simulation variables*



- loadsim.mi
- Set your variables in this file (sensors with prefix “s_”)
- loadmission loadsim.mi

```
sensor: s_ini_lat(deg) 4138.1060                # Ashumet pond
sensor: s_ini_lon(deg) -7032.0640               # Ashumet pond


sensor: s_water_depth_avg(m)             200.0  #master data default is 30
sensor: s_water_depth_delta(m)             0.0
sensor: s_water_depth_wavelength(m)      100.0

sensor: s_wind_speed(m/s)      9.0              # how fast the wind is blowing,
                                                # 3.0 ==> 5.4 knots
sensor: s_wind_direction(rad) 0.0               # Direction wind is blowing FROM

sensor: s_water_speed(m/s)       0.05           # Current speed,   0.5 ==> 1knot
sensor: s_water_direction(rad)  4.712           # direction current is going TO,
                                                # toward the west
```

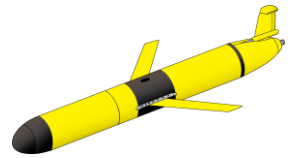
- Tips for Simulation:
  - You can modify autoexec.mi for simulation
  - Change the name? (unit_XXX_sim)
  - Uninstall iridium?
  - For pocket simulators, set the time & date

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Surface Drift



```
> put u_max_time_in_gliderdos 3600
> callback 30
```

Assign a callback-30 script on the iridium server

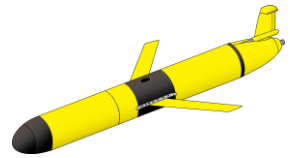
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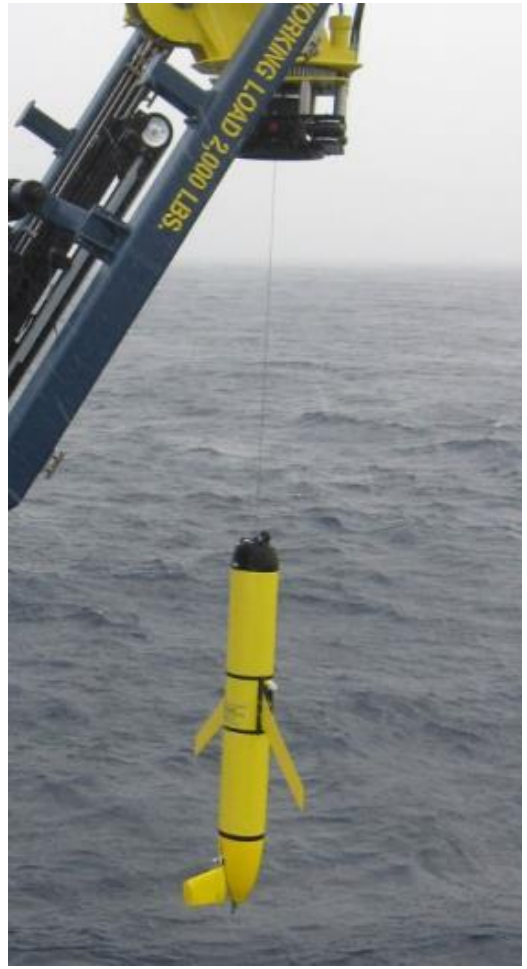
Nose Recovery System

– *this will cause the glider to burn the nose release!!!*



The command to burn the nose release wire to recover using the nose recovery system:

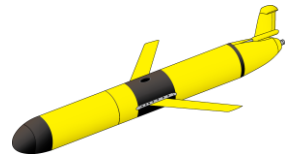
put c_recovery_on 1



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Notes - *Regarding Training & Qualified Personnel*



- **Only** trained and qualified personnel should operate and maintain the glider.
- Teledyne Webb Research conducts regular training sessions several times a year. Glider users should attend a training session and understand basic glider concepts and terminology.
- Contact glidersupport@teledyne.com for information regarding training sessions.
- Company policy is to fully support only properly trained individuals and groups.
- Only personnel who have attended a Teledyne Webb Research training session should use this document

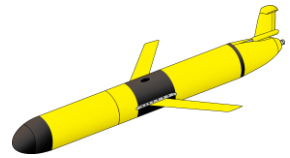


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Notes - *Regarding Training & Qualified Personnel*



Continued & Advanced Training

- Come back to another session in a year or two as a refresher and bring your intermediary questions
  - 2-4 sessions a year
- Watch for information about Advanced Training sessions in forum and by email
  - 1 a year

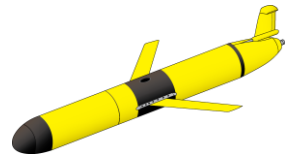


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Thank You!



We would like to thank you for attending this training session and look forward to assisting you as you deploy your gliders!

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